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FARMER'S ENTREPRENEURIAL MINDSET AND THE INTENTION TO ADOPT CLIMATE-SMART AGRICULTURE: THE CASE OF COFFEE FARMERS IN GREATER LUWERO, UGANDA

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ABSTRACT

Objective: The study aimed to explore the entrepreneurial mindset and the factors influencing smallholder coffee farmers' intentions to adopt or continue adopting climate-smart agriculture technologies with a case study in greater Luwero, Uganda.

Methods: A cross-sectional survey of 230 smallholder coffee farmers in greater Luwero, Uganda, formed the basis for the study. Data was collected using a questionnaire based on the Business model canvas, Theory of Planned Behavior, Entrepreneurial Orientation theory, and Diffusion of Innovation Theory. The study employed structural equation modelling to predict farmer attitudes and intentions to adopt or continue adopting climate-smart agriculture technologies and cluster analysis to group farmers based on their entrepreneurial mindset.

Results: Segmentation analysis identified four groups of coffee farmers; 20.4% enterprising, 40.9% indecisive, 25.2% conservative and 13.5% non-enterprising and they differ significantly based on farmer characteristics and compliance with business model canvas elements. The perceived innovation characteristics significantly predicted coffee farmers' attitudes toward adopting climate-smart agriculture technologies. While subjective norms constitute a potential barrier, attitudes, perceived behavioural control, and entrepreneurial orientation merged to positively and significantly influence coffee farmers' intention to adopt climate-smart agriculture technologies. On average, 80% of coffee farmers intend to adopt labour-intensive climate-smart agriculture technologies, compared to only 49.2% for capital-intensive technologies, of which 87% were enterprising coffee farmers.

Conclusion: The study's findings added to the literature regarding adopting agricultural technologies, notably entrepreneurial orientation, a novel variable in assessing farmers' intentions to adopt new technologies. The small percentage of enterprising farmers indicates a low level of entrepreneurism among coffee farmers hence the low intentions to adopt climate-smart agriculture technologies. Future studies should validate these findings and consider the multidimensional perspective of entrepreneurial orientation, use data gathered over time, look at actual adoption rates, and assess the impact of climate-smart agriculture technologies. Examining how farmer entrepreneurial mindset and perceptions influence climate-smart agriculture adoption intention offers a solid platform for more targeted scaling policies and intervention strategies to be designed and implemented in a more sustainable manner.

Keywords: Entrepreneurial; Adoption; Climate-Smart Agriculture; Diffusion of Innovation; Theory of Planned Behaviour; Uganda; Smallholder farmers; Business model

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DEDICATION

This thesis is a dedication to my family and friends, who are very influential in shaping my career.

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ACRONYMS

BMC	Business Model Canvas
CCAFS	Climate Change and Food Security
CGIAR	Consultative Group for International Agricultural Research
CIAT	International Center for Tropical Agriculture
CSA	Climate-Smart Agriculture
DOI	Diffusion Of Innovation
EO	Entrepreneurial Orientation
EU	European Union
FAO	Food and Agriculture Organisation of the United Nations
GHG	Green House Gas
GOU	Government Of Uganda
ICO	International Coffee Organisation
IITA	International Institute of Tropical Agriculture
IPCC	Intergovernmental Panel on Climate Change
IRB	Internal Review Board
ISO	International Organisation for Standardization
MAAIF	Ministry of Agriculture Animal Industry and Fisheries
NGO	Non-Governmental Organisation
OECD	Organisation for Economic Co-operation and Development
SDGs	Sustainable Development Goals
SSA	Sub-Saharan African
TPB	Theory of Planned Behaviour
TRA	Theory of Reasoned Action
UBOS	Uganda Bureau Of Statistics
UCDA	Uganda Coffee Development Authority
UNCCS	United Nations Climate Change Secretariat
USAID	United States Agency for International Development

CHAPTER ONE: INTRODUCTION

1.1 Background

In recent decades, climate change has become a global concern. High temperatures, irregular precipitation and high levels of greenhouse gases (GHGs) in the atmosphere were all noted in a recent UN Climate Change Secretariat (UNCCS) study (UNCC, 2019). The rising GHG levels trap heat in the atmosphere leading to the gradual warming of the earth's surface, oceans and atmosphere. Nevertheless, the scientific dispute over climate change continues, with natural systems and anthropogenic activities pronounced as the sources of GHG emissions in a ratio of 8:10, respectively (Edenhofer *et al.,* 2014; Yue & Gao, 2018; Yoro & Daramola, 2020). Anthropogenic activities are the leading emitters of GHGs. The Intergovernmental Panel on Climate Change (IPCC) noted that agriculture accounts for a significant share of anthropogenic emissions with its wide range of operations and frequent land-use changes (IPCC, 2014). Moreover, the Food and Agriculture Organisation of the United Nations (FAO) report observed that agricultural land use accounted for 17% of the worldwide GHG emissions from all sectors. The contribution increases substantially with land-use change (FAO, 2020; OECD, 2016).

Agriculture's contribution to overall emissions varies per region; for European Union (EU), because of its advanced technology, agriculture contributes 10% to the EU's overall GHG emissions, 8% in the United States and 11% in China (Richards *et al.,* 2015; Smith *et al*., 2014). Agriculture accounts for 35% of national GHG emissions in developing countries. The amounts vary widely between states, with Uganda's agriculture contributing 46%, slightly lower than the highest at 50% (Richards *et al*., 2015; USAID & CIAT, 2017). Nonetheless, researchers widely acknowledge that emissions from agriculture do not self-regulate, and countries should adopt strategies to lower them to meet the climate change target of limiting warming to 1.5 or 2 degrees Celsius over pre-industrial levels (Gernaat *et al*., 2015; Reisinger *et al*., 2013; Wollenberg *et al.,* 2016). Instead, if emissions are not regulated, they are deemed harmful to the terrestrial ecosystem and add to its pressure (Yue & Gao, 2018). Agriculture contributes significantly to the climate change problem, a threat to tropical countries.

Coffee remains the world's most popular beverage, an essential tropical export crop and a growth driver for economies in more than 40 countries (International Coffee Organisation (ICO), 2021; United States Department of Agriculture, 2021). Coffee is Uganda's leading agricultural commodity and the most significant contributor to the country's exports, valued at US$ 492 million, representing 16% of the total exports in 2017/2018 (Uganda Coffee Development Authority (UCDA), 2019). Besides being fundamental for foreign exchange generation, the Uganda coffee sector employs more than 12 million people throughout the value chain from production to processing, and it is the primary source of income for the rural poor inhabitants in over 30 districts in the country (Uganda Bureau Of Statistics (UBOS), 2018). Most coffee farmers are smallholders farming on less than two hectares of land and totalling 1.7 million coffee households (Bunn *et al*., 2019), growing two main varieties; Arabica and Robusta coffee. Three hundred fifty-three thousand nine hundred seven hectares (353,907) are under coffee production in Uganda, with coffee gardens ranging from 0.5 to 2 ha intercropped with bananas and other food crops (Mugoya, 2018). Coffee yields vary significantly between 573.3 to 2700 Kgs of clean, dry coffee/ha for traditional Robusta coffee depending on the production technologies (Wang *et al*., 2015). Despite being low, the current coffee yields in Uganda are also on a reducing trend (ICO, 2019; UBOS, 2020). As such, sustainable agricultural technologies are vital to stabilize and positively influence coffee yields, amidst climate change (MAAIF, 2015).

Due to its dependency on suitable conditions of sunlight, soil, heat and water, coffee has been demonstrated to be impacted by climate change, with negative impacts more common than positive ones (Lemma *et al.*, 2016; Lobell *et al.*, 2015; Müller, 2013; Ray *et al*., 2015; Saadi *et al*., 2015; IPCC, 2014). The nature and magnitude of the impact depend on the sector's climate sensitivity, the geographical profile of the country, crop type, and availability of adaptive infrastructure (Ali *et al*., 2017; Kukal & Irmak, 2018; IPCC, 2014). Climate change poses a slew of threats to coffee, including crop physiological disturbance (Wang *et al*., 2018), pest explosion and attack (Skendžić *et al.*, 2021), increased plant disease (Zayan, 2018), and increasing water stress (Zhao *et al.*, 2017). Eventually, it causes wilting and drooping of the crop, delayed flowering and lower bean weight significantly reducing coffee yields, posing a severe threat to global food production and jeopardising food security (Zhao *et al*., 2017). When this is combined with the continual depletion of natural resources, guaranteeing long-term sustainability becomes an extreme challenge (Nawaz *et al.*, 2019; Palacios *et al*., 2018; Yu *et al*., 2016), hence the slow progress toward achieving Sustainable Development Goals (SDG 13: Climate Action; SDG 12: responsible consumption and production; SDG 2: zero hunger; SDG 1: No poverty). The dilemma of fulfilling rising global food demand while lowering environmental impacts casts agriculture as both a problem and a potential mitigator of the worst effects of climate change.

Climate change mitigation and adaptation necessitates the implementation of several complementary policies aimed at lowering land-use change rates, improving agricultural efficiency, increasing carbon sequestration, and prioritizing the use of renewable energy sources (Sardar *et al*., 2021; Harvey *et al.*, 2014; Megevand *et al*., 2013; Zimmerer, 2013). As a result, a broader examination of technology mitigation strategies with possibilities for reducing GHG emissions is required (Fellmann *et al.*, 2018; OECD, 2016). More recently, sustainable farming practices, and Climate-Smart Agriculture (CSA), have been proposed as relatively novel farming techniques that could revolutionize agriculture, thus enlightening the future of food systems. CSA presents a valuable alternative to counteract agriculture’s contribution to climate change, mitigate the effects of climate change, and enhance climate resilience (de Pinto *et al*., 2020; FAO, 2013; FAO, 2016; Lipper *et al.*, 2014; Steward e*t al*., 2019).

The technologies and practices developed under the CSA program in developing countries include integrated soil fertility management, agroforestry systems, conservation agriculture, physical garden infrastructure (contours, terraces, stone bunds, grassed rivers), and enhanced crop varieties. (Ampaire *et al*., 2017; Brown *et al*., 2018; Verkaart *et al*., 2019; Blaser *et al.*, 2018; Pradhan & Ranjan, 2016). Additionally, Sub-Saharan African (SSA) countries have received significant international funding to develop legislative frameworks, legally binding procedures with norms, and incentives to strengthen CSA programs in their own countries (Blaser *et al*., 2018; Dinesh *et al*., 2017; Torquebiau *et al*., 2018; Weiler *et al*., 2018; Zougmoré et al., 2016). Such tools have significant potential for improving crop yields, better natural resource management, climate adaptation, and reduced environmental impact. However, the benefits of CSA can only be realized around the globe if farmers accept and adopt the technologies (FAO, 2020b). Farmers, as farm-level decision-makers, play a critical role in facilitating the transition to more climate-resilient coffee production (Bunn *et al*., 2019). Therefore, understanding farmers' adoption intentions are instructive in informing the government and other stakeholders about the enablers and barriers of CSA implementation, which is necessary for designing sustainable, scalable interventions.

Although different stakeholders have backed CSA research and development, the adoption of CSA technologies is still low, limiting its potential (Aggarwal *et al*., 2018; Makate *et al*., 2019; Karlsson *et al*., 2017; Sommer *et al*., 2018; Teshome *et al*., 2016). Sociologists and economists have all contributed to understanding farmers' decisions to adopt agricultural technologies. Earlier studies (Engel & Muller, 2016; FAO, 2016; Khatri-Chhetri *et al*., 2017; Kpadonou *et al.*, 2017) recognize that farmers' adoption of agricultural innovations depends on technological, behaviour and socioeconomic factors. Farmers in resource-poor developing nations lack sufficient access to fundamental production factors such as land, labour, and capital, hence it is risky for them to take up resource-intensive technologies (Ampaire *et al.,* 2017; Brown *et al*., 2018; Nakabugo *et al*., 2021; Paustian et al., 2016; Pradhan & Ranjan, 2016). On the other hand, farmers' adoption of technologies depends on technology benefits, training, demonstration and ease of use (Maina *et al*., 2020; Mmapatla *et al*., 2021). Furthermore, researchers have found that social norms, personal efficacy and perceived behavioural control play a significant role in adopting sustainable practices (Zeweld *et al*., 2017). Farmers’ networks and knowledge dynamics are critical adoption drivers in social science studies (Marra *et al.*, 2003). Moreover, farmers’ attitude toward technology is the strongest predictor of their intention to adopt sustainable agriculture practices (Chen, 2017; Van Hulst & Posthumus, 2016). So, increasing the positive attitude of farmers toward climate-smart agriculture technologies could be more effective in accelerating the adoption rate and sustainable scaling of the technologies (Jahanmir & Cavadas, 2018).

Accordingly, recent research links the adoption of novel technologies to the users' entrepreneurial mindset (Dias *et al*., 2019; Passarelli *et al*., 2021). Farmers who are innovative, proactive and risk-taking adopt agricultural technologies more quickly than their counterparts (Pindado & Sánchez, 2017; Suvanto *et al*., 2020). Furthermore, knowledge of the technologies' strategic incentives and value configurations influences farmers' decision to use technologies in their farming business (Bogataj Habjan & Pucihar, 2017; Boons & Lüdeke-Freund, 2013; Tageo *et al*., 2020). However, entrepreneurship research has predominantly neglected agricultural enterprises, and the influence of entrepreneurial orientation on farmers’ adoption intentions is still mostly unknown (Dias *et al*., 2019).

Like any other agricultural technology, CSA implementation occurs at the farm level, where farmers make the decisions; hence, it is crucial to understand farmers' perceptions, reactions and the rationale to adopt CSA (De Rosa *et al*., 2019). Besides, agricultural economists must investigate the entrepreneurial mindsets of farmers because they are a means of overcoming the financial obstacles that hinder adoption by resource-constrained smallholder farmers. As such, the Diffusion of Innovation and Theory of Planned Behaviour was used in this study to understand better the technological and behavioural aspects that influence farmers' intentions to adopt and continue adopting CSA. Similarly, we used the Business Model Canvas and the Entrepreneurial Orientation theory to investigate farmers' entrepreneurial mindsets and add a valuable factor for predicting farmers' innovation uptake to the existing literature. By informing about the enablers and barriers of CSA adoption, which is appropriate for policy support, the study adds to the expanding body of research on the targeted sustainable scaling of CSA technologies.

1.2 Problem Statement and Rationale

Coffee is imperative in the economic development of Uganda, and we cannot overlook its contribution to poverty reduction. Over the years, the country has experienced slower and fluctuating growth in coffee volumes, with increased cultivated land. In 2013, the Uganda Bureau of Statistics (UBOS) reported 236,000 tons of coffee produced, which decreased to 210,000 tons in 2014 (UBOS, 2018). The volume rose to 236,000 tons in 2015 and 244,000 tons in 2016 (UBOS, 2018). The total amount of coffee produced in 2017 was 303,000 tons, dropping to 285,000 tons in 2018 and increasing to 313,000 tons in 2019 (UBOS, 2019; UBOS, 2020). The instability in coffee production is attributed to poor agricultural land management practices, limited input use, environmental degradation, and climate change, as evidenced by erratic weather patterns and increased season variability (FAO, 2016; Tumwine *et al*., 2019; Nsubuga *et al*., 2021; Twinomuhangi *et al*., 2021; Wichern *et al*., 2019). Climate change has enormous impacts on coffee production; specifically, high temperatures and irregular precipitation enhance coffee's susceptibility to pests and diseases (Adhikari *et al.*, 2020). This effect affects flowering, and fruiting (Chacón-Villalobos *et al.*, 2021) hence declines in coffee yield and loss of coffee-optimal areas (Pham *et al.*, 2019; Tavares *et al.*, 2018). The low and fluctuating coffee yields produced by coffee farmers in Uganda significantly impede the sector's growth, endangering farmers' livelihoods and jeopardizing food security across the country. Sustainable technologies, specifically CSA, are necessary to reverse Uganda's declining coffee production trend.

CSA technologies have been proposed as a farming approach to promote resilience against climate change and enhance sustainable coffee production (De Pintoid *et al*., 2020; FAO, 2013; FAO, 2016; Lipper *et al*., 2014; Steward *et al*., 2019). In the case of food security, Sun *et al.* (2018) affirm that due to the adoption of CSA, winter wheat yields grew by 36.5% in China from 1981 to 2010. Similarly, in Pakistan, Sardar *et al*. (2020) observed that farmers that adopted CSA technologies had greater yields of 32% and 44% for cotton and grain, respectively, and higher farm income of 45% for rice and 48% for wheat compared to the non-adopters. Implementing multiple CSA technologies withstands climate change's impacts on Ethiopia's production and diversity of Arabica coffee (Asegid, 2020). Implementing agroforestry pruning, alley cropping, and silvopasture in Kenya increased coffee productivity by 24%, 21% and 195%, respectively (Lamanna *et al*., 2020). The benefits of CSA in terms of mitigation and adaptation are recognized, but scientists rarely evaluate them because carbon monitoring is expensive and uncertain (Kearney *et al*., 2017). In an experiment conducted in Indonesian rice fields, CSA lowered GHG emissions by 7-23% of Global warming potential compared to non-CSA farmer practices (Ariani *et al.*, 2018). Climate change and variability will not, without a doubt, diminish environmental quality, agricultural output and food security in Sub-Saharan Africa if opportunities for CSA technologies and enabling policies are encouraged and embraced by governments and farmers (Zougmoré *et al*., 2018).

Fortunately, the Government of Uganda (GOU), through the Ministry of Agriculture Animal Industry and Fisheries (MAAIF), adopted CSA and formed the National Climate-Smart Agriculture Program (MAAIF, 2015) with a vision to counteract climate change. Similarly, commercial enterprises, research institutions, and non-governmental organizations (NGOs) have created and supported several CSA technologies to aid climate change adaptation and food security. Notably, the Climate Smart Investment Pathway (Stepwise), a breakdown of the best farming practices into 18 CSA technologies, was developed by the International Institute of Tropical Agriculture (IITA) and deployed to coffee growing regions of greater Luwero (Jassogne *et al*., 2017). These technologies include land and water resource management, soil and water conservation, planting shade trees, and intercropping coffee and food crops, among others (Faisal *et al.,* 2021; GOU, 2018; Jassogne *et al*., 2017; Nakabugo *et al*., 2019; USAID & CIAT, 2017). These technologies can revitalize agriculture sustainably and productively while addressing various societal and environmental demands (FAO, 2020).

Unfortunately, CSA technologies that are known to be sustainable and productive have mainly remained unused with a relatively low rate of adoption in developing countries, which has little impact on natural resource management sustainability or climate resilience (Brown *et al*., 2018; Paustian *et al.*, 2016; Pradhan & Ranjan, 2016; GOU, 2018). Ampaire *et al*. (2017) observed a slow implementation of CSA policies due to limited technical capacity and lack of resources at the community level, which widens the agricultural extension gap. Additionally, limited financial services and agrarian risks are significant impediments to adopting CSA technologies in Uganda (Bunn *et al.*, 2019; USAID & CIAT, 2017). Different factors influence farmers' decision-making processes regarding agricultural innovations, but the literature on the determinants of CSA technology adoption at the farm level is inconclusive. Researchers have primarily used the Diffusion of Innovations (DOI) theory to understand the factors underlying the adoption of agricultural technologies (Rogers, 2003), but also behavioural models, particularly the Theory of Planned Behaviour (TPB), have been employed (Ajzen, 1991). Both theories have been used extensively and proven successful in predicting farmers' motivation toward adopting technologies. Meanwhile, the combined use of DOI and TPB has been found complementary and offers additional explanatory power regarding the adoption decision (Jung Moon, 2020; Warner *et al*., 2021; Weigel *et al*., 2014) though there is limited literature about farmers. The study proposes a combination of DOI and TPB constructs to influence coffee farmers' attitude and intention to adopt or continue adopting CSA technologies respectively.

Recently, entrepreneurial orientation has gained attention as a compelling belief to increase the adoption of technologies (Dias *et al*., 2019; Long *et al.*, 2017; Passarelli *et al*., 2021). The farmer's configuration of business operations depicts the intrinsic desire and orientation to engage in entrepreneurial activities (Joyce & Paquin, 2016; Rosenstock *et al*., 2020; Teece, 2018). Suvanto *et al*. (2020) affirm that farmers' entrepreneurial identity increases the adoption intentions of protein-rich crops in Europe. As such, having an entrepreneurial mindset can help farmers adopt or continue adopting CSA technology and accelerate the transition to more sustainable agriculture (Passarelli *et al.*, 2021). However, literature is scarce regarding farmers' entrepreneurial orientation and Business Model conformity which this study intends to address. This study will leverage the Business Model Canvas and the Entrepreneurial Orientation theory to understand the entrepreneurial mindset of smallhoilder coffee farmers and how it affects their intentions to adopt or continue adopting CSA technologies. Such Information is important for assessing farmers' dynamic capacities and their ability to compete in a competitive context. It is also critical for developing appropriate policy interventions and targeted strategies for sustainably scaling the adoption of CSA technologies.

1.3 Objectives of the study

1.3.1 General Objective

To explore the entrepreneurial mindset of smallholder coffee farmers in greater Luwero, Uganda and the factors influencing their intentions to adopt or continue adopting climate-smart agriculture technologies.

1.3.2 Specific Objectives

1. To analyze farmers' entrepreneurial orientation and compliance with the business model canvas elements in their farms.
2. To determine the influence of perceived technological attributes on farmers' attitudes towards adopting climate-smart agriculture.
3. To evaluate the influence of entrepreneurial orientation and psychological factors on the farmer's intention to adopt or continue adopting climate-smart agriculture.

1.3.3 Research Questions

1. How do entrepreneurial orientation and compliance with the business model canvas elements vary amongst coffee farmers?
2. What is the influence of perceived technological attributes on farmers' attitudes towards adopting climate-smart agriculture?
3. What influence do entrepreneurial orientation and psychological factors have on farmers' intention to adopt or continue adopting climate-smart agriculture technologies?

1.4 Organisation of the paper

The remaining sections of the thesis flow as follows. First, chapter two goes over an empirical examination of climate-smart agriculture, farmers' attitudes towards CSA, agricultural technology adoption by farmers and farmers' entrepreneurial mindsets. To present the theories that underpin the study, we explore the Entrepreneurial Orientation, the Business Model Canvas, the Diffusion of Innovation, and the Theory of Planned Behaviour. In chapter three, we present the methodology employed in answering the research questions, while chapter four comprises the presentation of the findings of the study. Finally, the last chapter (chapter five) discusses the results, study limitations, suggestions for future research, significance of the study and the conclusion.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter concentrated on scrutinizing literature related to CSA, farmers' attitudes and adoption of CSA technologies, farmers' conformity to the BMC and its effect on adoption of CSA technologies, their critiques, research gaps and a conceptual framework that described the correlation amidst research variables. Additionally, it addresses the theories that anchor this study.

2.2 Empirical Literature Review

2.2.1 Climate-Smart Agriculture

The term CSA was first used in debates regarding climate change and agriculture by the FAO (2009). A year later, the CSA concept was defined as agriculture with three goals: increasing production to support increased revenues, food security, and development; building climate resilience and reducing GHG emissions; and improving carbon sequestration (FAO, 2010). More precisely, CSA practices are agricultural techniques with the potential to improve resource efficiency while enhancing the resilience of farming systems and the people who rely on them (FAO, 2013). CSA is a promising approach representing a valuable alternative to counteract agriculture's contribution to climate change, mitigate climate change effects and enhance climate resilience (de Pinto *et al*., 2020; FAO, 2013; FAO, 2016; Lipper *et al.*, 2014; Steward e*t al*., 2019). Since then, CSA has served as a focal point for global investment in agricultural innovation research aimed at developing sustainable farming technologies and practices that allow agriculture to operate in an environmentally and natural resource-conserving way (Chandra *et al*., 2017; FAO, 2016; Sommer *et al*., 2013; Teklewold *et al*., 2017; Thierfelder *et al*., 2017). Fortunately, international institutions such as the FAO, World Bank, local and international NGOs, and the Consultative Group for International Agricultural Research (CGIAR) Climate Change and Food Security (CCAFS) program launched CSA projects at the country and regional levels (Lipper *et al*., 2018).

Following the World Bank and CCAFS program's development of "country CSA profiles," multiple CSA technologies have been identified and developed for various countries based on their conditions and contexts (Lipper *et al*., 2018). Generally, technologies advocated for developing countries include water conservation, soil fertility management, agroforestry systems, conservation agriculture, physical garden infrastructure (contours, terraces, stone bunds, grassed rivers), and enhanced crop varieties. (Ampaire *et al*., 2017; Brown *et al*., 2018; Verkaart *et al*., 2019; Blaser *et al.*, 2018; Pradhan & Ranjan, 2016). The Uganda CSA country profile classifies a practice as CSA if it improves food security and at least one other goal (adaptation or mitigation). USAID & CIAT (2017) selected different technologies for specific production systems. In the case of Climate-Smart coffee production, the International Institute for Tropical Agriculture established 18 technologies for coffee under the Climate Smart Investment Pathway (Stepwise) (Jassogne *et al*., 2017). These technologies include good agronomic practices like mechanical weeding, pruning, de-suckering, gap filling, manure application, mulching, inorganic fertilizers, chemical pesticides, shade tree planting and management, intercropping, cover crops, cultural and chemical pest control, irrigation, and soil and water conservation structures, among others. Although governments, businesses, and non-governmental organizations have backed CSA research and development, adoption of CSA technology and techniques is still low, and its potential has yet to be fully realized (Aggarwal *et al*., 2018; Makate *et al*., 2019; Karlsson *et al*., 2017; Sommer *et al*., 2018; Teshome et al., 2016). Resource restrictions are a major stumbling factor in deploying CSA technology in several studies (Engel & Muller, 2016; FAO, 2016; Khatri-Chhetri *et al*., 2017; Kpadonou *et al.*, 2017). However, there is still limited evidence to ascertain the factors influencing farmers' intentions to adopt and continue adopting CSA technologies. The study focussed on 18 CSA technologies in the Climate Smart Investment Pathway (Stepwise) developed by the International Institute for Tropical Agriculture (Jassogne *et al*., 2017) for coffee production in Uganda to answer the research questions.

2.2.2 Farmers' attitude towards Climate-Smart Agriculture technologies

Farmers' attitude towards technology is the most significant factor influencing behavioral intention to adopt and continue using agricultural technologies (Chen, 2017; Dalila *et al*., 2020; Hasan & Suciarto, 2020; Prati *et al*., 2012). Attitudes are personal feelings about a specific behaviour that can be positive, negative or neutral (Fishbein & Ajzen, 1975; Ajzen & Albarracín, 2007). Several studies noted a significant and positive influence of attitude on behavioral intention to adopt technologies (Cristea & Gheorghiu, 2016; Dalila *et al*., 2020; Hasan & Suciarto, 2020; Mead & Irish, 2021; Shan *et al*., 2020), but also as a mediator of the relationship between technology adoption and subjective norms (Kim *et al.*, 2013). As such, increasing the positive attitude of farmers toward climate-smart agriculture technologies could be more effective to accelerate the rate of adoption and sustainable scaling of climate-smart agriculture technologies (Jahanmir & Cavadas, 2018).

Fishbein & Ajzen (1975) assert that attitudes are determined by the strength of behavioural beliefs regarding the potential outcomes of the behaviour which predicts likelihood of adoption. Accordingly, Tsai *et al*. (2019) affirm that perceived usefulness, compatibility and transition costs significantly determine users' attitudes towards adoption of telehealth innovation. Similarly, among the factors that influence the physicians' attitude toward using a smart phone were the innovation characteristics of compatibility, job relevance, the internal environment, observability, personal experience, and the external environment (Putzer & Park, 2012). The technology acceptance model (TAM) was used by Porter & Donthu (2006) to determine which consumer beliefs account for attitudes regarding internet usage. It was discovered that users' attitudes about using the internet are positively influenced by perceived; usefulness, ease of use, and access barriers. Jung Moon (2020) also found that attitude towards the adoption of electric vehicles was significantly predicted by relative advantage, compatibility and observability. Although studies have focused a lot on how users perceive innovations, other potential responses to adoption, such as the factors that influence farmers' attitudes toward adopting climate-smart agriculture technologies, have not been properly examined. Thus, this study explores the influence of the diffusion of innovation's perceived technology characteristics on smallholder coffee farmers' attitude towards adoption of CSA.

2.2.3 Farmers' adoption of Climate-Smart Agriculture technologies

Farmers' adoption of CSA technologies is influenced by several factors that guide the study's analysis. A comprehensive review of the literature on adopting agricultural technologies demonstrates the predominance of farm-related and socio-economic factors. (Engel & Muller, 2016; FAO, 2016; Khatri-Chhetri *et al*., 2017; Kpadonou *et al.*, 2017). Farmers operate in complex dynamic socioeconomic realities and characteristics that influence their decisions regarding CSA technology adoption (Makate *et al.*, 2018; Sardar *et al*., 2020). Mmapatla *et al*. (2021) affirm that gender, access to finance and marital status influenced farmers' choice to adopt CSA technologies.

Nevertheless, when farmers encounter alternative employment opportunities outside of agriculture, they ignore CSA technologies because they have alternative sources of income. In Nigeria, Onyeneke *et al*. (2018) assessed the status of CSA. Study findings revealed that farmers' education, income, farming experience, the land area cultivated, risk orientation of the farmer, gender, land ownership, household size, and mass media exposure are significant determinants of CSA adoption. More recently, Tran *et al*. (2020) also found similar results regarding farmers' adoption of CSA technologies in rice production in Vietnam. Contrarily, the age of the household head, the total land area held, petty trading, and formal employment reduced the likelihood of adopting more than two CSA strategies in Southern Malawi (Maguza-Tembo *et al*., 2017). Moreover, study scores noted that CSA awareness and training increased the likelihood of adopting CSA technologies. This factor emphasizes the importance of information flow and communication from technology providers to end-users, influencing perceptions of the technology.

Meanwhile, farmers' perception of CSA technology in the adoption decision process is widely recognised, although characteristics of CSA technologies are rarely considered (Jellason *et al*., 2020; Sardar *et al.*, 2020; Weniga Anuga *et al*., 2013). A study by Mmapatla *et al*. (2021) utilized a multinomial logistic regression model to examine and affirm that training and demonstration, technology requirements and benefits related to hybrid seeds are significant influencers of smallholder farmers' adoption of CSA in South Africa. Maina *et al*. (2020) also pointed out that the perceived benefits of brachiaria grass significantly influenced its adoption among dairy farmers in Kenya's Eastern and Western regions. While perceived usefulness plays a significant role in adopting sustainable practices in Ethiopia, social norms, personal efficacy, and training are equally important predictors (Zeweld *et al*., 2017). This result validates behavioural scientists' claims that subjective emotions and social norms also determine human thoughts, relatively not pure logic as expected by classical economic theories.

Even though there has been limited integration of psychological constructs in agricultural adoption literature, some studies have leveraged behavioural constructs to explain farmers' intention to adopt CSA technologies. A survey of 422 dry land farmers in Australia revealed that values, attitudes, and norms are significant predictors of adopting complex land management practices (Price & Leviston, 2014). Similarly, strong evidence was found that attitude and perceived behavioural control influence farmers' intentions to adopt conservation agriculture practices in Kenya (Van Hulst & Posthumus, 2016), moreover, experimentation and learning were proven essential to support adoption as they contribute both to realistic attitudes and an improved perceived behavioural control. As such, research should explore psychological factors to provide a more accurate estimate of the determinants of farmers' intention to adopt CSA technologies.

2.2.4 Farmers' Entrepreneurial Mindset

Recently, farmers' roles have gradually shifted from merely being food producers to innovative and risk-taking business owners capable of operating in a competitive environment (Kangogo *et al*., 2021; Suvanto *et al*., 2020). Apart from that, farmers seeking autonomy tend to overcome the challenge of production factor constraints and limited access to formal funding (Nakabugo *et al*., 2021) by depending more on their labour, skills, and knowledge, as well as possible cooperation from family and social networks (Milone & Ventura, 2019). Combined with a specific focus on the main aspects of their business operations, this resets the imbalances between internal and external resources, allowing farmers to operate on their own in ways considered strategic for the farm's success (Zarazua de Rubens *et al*., 2020). Autonomy, innovation, risk-taking, collaborations, networks, and the understanding of business configuration are among the entrepreneurial value-sets that allow an individual to have a distinct entrepreneurial mindset (Khatri-Chhetri *et al*., 2017; Kpadonou *et al.*, 2017; Pindado *et al*., 2018).

According to Kirkley (2016), an entrepreneurial mindset is a particular collection of values (beliefs) and requirements that equip an individual with the intrinsic desire and orientation to engage in business activities. It is also an innovative technique for creating and exploiting business chances, whether employed or running own enterprise (Mazzarol & Reboud, 2020). According to the literature, entrepreneurial mindset and pursuit depend on the entrepreneur's internal knowledge and ability to acquire and exploit information from various sources (Kirkley, 2016). Although its use is limited in agricultural research, the concept is familiar in entrepreneurship research with its roots. Nonetheless, a growing body of research has examined the methods by which farmers participate in entrepreneurial activities over the last decade (Mazzarol & Reboud, 2020; Milone & Ventura, 2019; Suvanto *et al*., 2020). Some authors have linked the notion to farmers' development of non-agricultural businesses (Pindado & Sánchez, 2017a). Other authors have highlighted farmers' participation in entrepreneurial opportunities along the agricultural chain, such as developing new products and innovations in farming and marketing practices (Suvanto *et al*., 2020). Both perspectives originate from entrepreneurialism, a process in which individuals take advantage of favourable market conditions to develop and expand new business ideas (Dias *et al*., 2019). This process takes two forms: the occupational and the behavioural. The former corresponds to owning and strategically operating a business (business model), while the latter corresponds to entrepreneurial orientation in the context of capturing an economic opportunity (Pindado & Sánchez, 2017a). While more studies address farmers' entrepreneurial mindsets, particularly entrepreneurial orientation and business models, our understanding of these concepts in the context of agricultural technology adoption is still limited.

Although entrepreneurial orientation is a relatively new concept in agriculture, it has gained much attention as a tool for evaluating farmers' entrepreneurial mindset (Dias *et al*., 2019). Several studies have examined entrepreneurial orientation based on the three dimensions of; risk-taking, innovativeness and growth orientation (Pindado & Sánchez, 2017; Suvanto *et al*., 2020). Jassogne *et al*. (2017) revealed that the level of farmers' entrepreneurism varied based on age, experience, and asset control, among others. As a managerial attitude, entrepreneurial orientation reflects in the strategic business operations that add value to the firm (business model). A business model describes the firm's value configurations in economic, social, and cultural contexts and is essential to entrepreneurial management (Boons & Lüdeke-Freund, 2013). Specifically, the Business Model Canvas (BMC) influences the firm's dynamic capabilities and informs about the feasibility of an investment strategy (Joyce & Paquin, 2016; Teece, 2018). Following Osterwalder & Pigneur (2010), we conceive BMC as a set of nine key elements: revenues, cost structures, value proposition, channels, key resources, customer relations, customer segments, key partners and key activities. Metallo *et al.* (2018) revealed that entrepreneurs do not have the same relevance to all BMC elements. In the internet of things industry, Metallo *et al.* (2018) found that the most relevant elements to the perceived users and providers are; Key partners, key resources, value propositions and customer segments. Similary, Müller (2019) affirm that key resources, value propositions, key partners, customer segments and customer relations are the most cited BMC blocks by the small and medium enterprises in German. To conclude, we see an entrepreneurial mindset as a multifaceted construct that integrates multiple dimensions of entrepreneurial orientation and compliance with BMC elements. Nevertheless, to the best of our knowledge, there is no literature regarding the perceived importance of BMC elements in the agricultural industry.

2.3 Theoretical Literature Review

The theories that ground this study include; Entrepreneurial Orientation (EO), Business Model Canvas (BMC), Diffusion of Innovation Theory (DOI), and Theory of Planned Behavior (TPB). EO focuses on the characteristics of an entrepreneur, including risk-taking, innovativeness, and proactiveness. The BMC visualises the entrepreneur's business operations under nine building blocks: revenues, cost structures, value proposition, channels, key resources, customer relations, customer segments, key partners, and key activities. relative advantage, compatibility, complexity, trialability, and observability are technology attributes that influence adoption and are components of the DOI theory. The Theory of Planned Behaviour (TPB) is concerned with behavioural variables, precisely attitudes, subjective norms and perceived behavioural control that perform a crucial role in forming behaviour. Thus, examining these models provides an opportunity to understand better the intention to adopt CSA technologies by coffee farmers in greater Luweero, Uganda.

2.3.1 Entrepreneurial Orientation (EO)

Entrepreneurial orientation (EO) is a well-established construct in entrepreneurship and management research. However, its popularity in agriculture research has only recently increased but it had previously been silent (Pindado *et al.*, 2018; Suvanto *et al.*, 2020). The idea has gained focus due to the growing need to understand farmers' strategic decisions (Miller & Le Breton-Miller, 2011). Unlike other entrepreneurs, farmers operate in an unstable environmental (climate variability) and economic (production factor constraints) situation, so their entrepreneurial orientation is crucial to enhance their survivability and adaptation (Pindado *et al*., 2018). One of the features of farm enterprises in the agricultural sector is a sole proprietorship, which implies that the individual farmer is the primary participant in operational decisions and choices made on the farm (Milone & Ventura, 2019; Pindado *et al*., 2018). As such, the study concentrated solely on the entrepreneurial orientation of individual farmers. The first conceptualization of EO was by Miller (1983) as a three-dimensional view of constructs of innovativeness, risk-taking, and proactiveness. Lumpkin & Dess (1996) expanded EO seven years later by adding two new dimensions: competitive aggressiveness and autonomy, resulting in a five-dimensional version of EO. Since then, the two prominent notions have coexisted in the literature, each offering unique insights; however, researchers have frequently relied on Miller's (1983) three-dimensional conceptualization for accumulating knowledge around a stabilized concept (George & Marino, 2011). EO hypothesizes that the individual entrepreneurial orientation is a function of their risk taking, innovativeness and proactiveness (Miller, 1983; Lumpkin & Dess, 1996).

Risk-taking is the entrepreneurs' willingness to undertake risky resource commitments to attain specific goals (Miller, 1983). Entrepreneurs are known as "risk-takers," and the fear of failure has a detrimental impact on the decision to start a firm (Dias *et al*., 2019). According to EO theory, strategic risk includes stepping into the unknown, devoting many resources to a venture, and substantially borrowing (Vora *et al*., 2012). Nonetheless, it is critical to understand that risk-taking is accepting calculated risk to gain rewards, as opposed to reckless betting with hardly any regard for the risks involved (Lumpkin & Dess, 1996; Miller & Le Breton-Miller, 2011). Because the farm and environmental gains are unpredictable, information and expertise about CSA are typically low, and failure means financial losses for farms; the study suggests that risk-taking affects CSA adoption. Moreover, the recent CSA country profile report emphasizes risks and financial constraints as barriers to farmers' adoption of CSA technologies (USAID & CIAT, 2017). However, Practising CSA has the potential to minimize agricultural risks by improving the resilience of farming systems and the communities (FAO, 2013).

Innovativeness is the ability of an entrepreneur to generate new ideas, uncover new market opportunities, and engage in creative processes that result in product, market, or technology advancements (Lumpkin & Dess, 1996). Innovativeness can imply anything from a desire to try out new products or services to a resolve to stay on the leading edge of technology, going beyond the existing state of the art (Lumpkin & Dess, 1996). Vora *et al.* (2012) argue that problem-solving and creative solutions demonstrate innovativeness and develop new products and services. It also includes the encouragement of new ideas and experimentation. As such, the methods for developing or adopting new products, services, or procedures are thus considered innovative. As is well known, innovativeness boosts a firm's potential growth and odds of survival in small businesses such as coffee farms (George & Marino, 2011; Miller, 1983). Farmers' eagerness to use CSA approaches, we believe, is a result of their innovativeness. For instance, individual farmers' desires to conduct environmentally friendly agriculture and gradually improve methods to increase the output to inputs ratio (Suvanto *et al*., 2020) demonstrate their innovativeness.

Being a first-mover and thus maintaining a competitive edge by identifying future opportunities is proactiveness. Proactiveness also means to act in advance of a future situation, tending to initiate change rather than reacting to change (Lumpkin & Dess, 1996; Miller, 1983). The future-oriented mindset is displayed by attention to emerging trends in consumer behaviour, market needs, and preferences, maximizing market opportunities (Lumpkin & Dess, 1996; Miller, 1983; Miller & Le Breton-Miller, 2011). However, this is not the case for farm businesses, which are rarely influenced directly by consumers or the processing industry that buys farmers' produce (Suvanto *et al*., 2020). A proactive farmer establishes new technology as the standard based on current trends (Milone & Ventura, 2019), such as CSA technologies, which respond to the agricultural industry's sustainability objective. As such, proactiveness influences farmers' innovativeness and willingness to take risks, impacting farm performance in terms of growth and profitability (Suvanto *et al*., 2020). Therefore proactiveness, risk-taking and innovativeness are crucial in forming farmers' entrepreneurial mindset. This study proposes a relationship between EO variables and coffee farmers' intention to adopt CSA technologies in greater Luweero, Uganda.

2.3.2 Business Model Canvas (BMC)

Business models have been integral to market research and trade for millennia to represent the venture's entrepreneurial strategy (Strupeit & Palm, 2016). However, their inclusion in agricultural research is more recent in assessing farm business. Business models are a business tool for explaining the creation, supply and capturing of value for customers, mechanisms to pay for value, and turn payables into profits (Teece, 2018). Business models draw on resource-based view theory, strategic network theory, cooperative strategies, transaction cost economics, and industrial organization strategy to orient the central focus of entrepreneurial strategy (Osterwalder *et al*., 2005; Strupeit & Palm, 2016). Developed by Osterwalder *et al.* (2005), the Business Model Canvas (BMC) is a fundamental tool that helps entrepreneurs understand their company's business model (Joyce & Paquin, 2016). Specifically, the BMC comprises nine building blocks: revenues, cost structures, value proposition, channels, key resources, customer relations, customer segments, key partners and key activities (Scaramuzzi *et al*., 2020). Besides representing a company's value configuration, the BMC was recently included in adoption studies to explain the uptake and scaling of technology in industrial sectors (Habjan & Pucihar, 2017; Strupeit & Palm, 2016). Most business and entrepreneurship management researchers emphasize that sticking to strategic business models characterizes successful businesses (Bogataj Habjan & Pucihar, 2017; Cortimiglia *et al*., 2016). Benefits derived from business models fall into five broad categories: strategic planning, business control and consistency, operational efficiency, understanding competition, information sharing and communication (Boons & Lüdeke-Freund, 2013; Liao *et al*., 2018; Sort & Nielsen, 2018).

The BMC provides a platform for identifying essential aspects or categories of business operations, analyzing the attributes and dynamics of those elements and the nature of interdependencies between them (Rydehell & Isaksson, 2019). According to Teece (2018), when the entrepreneur properly understands the business model, the alignment between high-level strategies and underlying activities of the company becomes more transparent. Assessing the knowledge and relevance of the BMC elements in an enterprise is crucial to gaining insights into the organisation's entrepreneurship management (Boons & Lüdeke-Freund, 2013). Furthermore, compliance with BMC elements guides entrepreneurs' perceptions and beliefs about what constitutes an appropriate business model, influencing their actions and business operations (Rydehell & Isaksson, 2019). For example, business models with external supporting actions increase the adoption and diffusion of technologies (Long *et al*., 2017). In the same vein, several authors, for instance, Rosenstock *et al*. (2020), found that by providing access to resources and a space for representation in decision making, inclusive and collaborative business models create value and assist the poor farmers to access the technologies.

Furthermore, small and medium enterprises' business models offering scientific and practical evidence of CSA technologies, good partnerships, good customer relationships and effective channels enabled CSA upscaling in Punjab, India (Groot *et al*., 2019). Thus, this study aims to provide light on business models as cognitive instruments coffee farmers use to arrange business activities to generate, deliver, and capture value.

2.3.3 Diffusion of Innovation Theory and adoption

The Diffusion of innovations theory (DOI) came out with Rogers (1962) in the communication to explain how, over time, an innovation gains momentum and diffuses through the population. The outcome of this diffusion is the adoption of the innovation, which implies that a person performs something different from the previous. For further clarity, Rogers (1995) defines innovation as an idea, practice, or object perceived as new by a unit of adoption. By this, we consider CSA technologies innovations as long as they are new to the farmer. DOI assesses how innovation is infused within a population and explains what happens as populations adopt and continue adopting it (Kaminski, 2021; Rambocas & Arjoon, 2012). In further clarifying his model, Rogers (2003) characterizes adopting innovation as a decision-making process to use innovation or not. On this note, it is evident that the adoption of a new idea, behaviour, or product does not occur concurrently in a society (H. Tran *et al*., 2017) but rather through a decision process that involves an evaluation of the innovation itself (Rogers, 2003).

Accordingly, DOI recognises five stages through which the innovation-decision process proceeds. Firstly, the initial knowledge of the innovation creates awareness, then persuasion by forming a favourable attitude about the innovation. From persuasion, the next step is deciding after considering the merits and demerits of an innovation. The fourth stage is implementation, where a person adopts and uses the innovation and lastly, confirmation, where a person decides whether to continue utilizing the innovation or not based on their personal experience (Rogers, 2003). The first two stages relate to the user's adoption behaviour and develop an attitude toward the innovation (Chou *et al*., 2012; Jung Moon, 2020). Jansson (2011) urges that the persuasion to form a favourable attitude about innovation results from the characteristics and personality traits. Within this process, innovation characteristics are vital to informing the attitude towards the technology and later the adoption decision. DOI posits that innovations offering more relative advantage, compatibility, complexity, trialability, and observability will be adopted faster than their counterparts (Izuagbe *et al*., 2016; Hameed & Counsell, 2014). He further reports that these five attributes explain 49-87% of the variance in the adoption rate of innovations. This study has developed a model including these five perceived innovation characteristics by Rogers (2003) to predict farmers' attitudes towards the adoption of CSA technologies.

Relative advantage indicates the degree to which an innovation is better than the idea, product, or program it supersedes. It is measured primarily by assessing an innovation's overall convenience and economic, social and technical superiority (Tan & Teo, 1998; Polatoglu & Ekin, 2003). Relative advantage is amongst the best predictors and is positively related to attitude toward innovation (Jung Moon, 2020) and innovation's rate of adoption (Rogers, 2003; Weigel *et al*., 2014). Putzer & Park (2012) also noted that smartphone relevance to the physicians' job was a strong determinant of attitude to use a smartphone. CSA technologies have the potential to increase coffee productivity, enhance resilience to climate risk and reduce GHGs. These benefits are hypothesised to influence farmers' attitudes positively.

Compatibility indicates the extent to which an innovation is consistent with the potential adopters' existing values, experiences, and needs. Tornatzky & Klein (1982) found that innovation is more likely to be adopted when it is compatible with individuals' job responsibilities and value systems. Compatibility of innovation with an initial idea positively shapes attitudes toward the innovation, as observed in mobile technologies (Putzer & Park, 2012) and the adoption of green practices in the restaurant industry (Chou *et al*., 2012). Mairura *et al*. (2016) argue that Individuals or organizations are more likely to adopt an innovation if it does not drastically disturb their lifestyle. Hence, CSA technologies must align with the coffee farmers' current values and experiences. The more compatible CSA technologies are with coffee farmers, the more positive the attitudes towards them reflect the adoption intentions. Complexity is the degree to which an innovation is perceived as relatively difficult to understand and use (Rogers, 2003). The more complex innovation is the lower its adoption rate (Tornatzky & Klein, 1982). Black *et al*. (2001) observed that a complicated and challenging service takes much more time to win over consumers. On this note, Jung Moon (2020) affirmed that the less complexity of the use of electric vehicles had a positive influence on the consumers' attitudes.

Trialability is the extent to pilot an innovation before a commitment to adoption. New ideas tested on an instalment plan are generally adopted more rapidly than innovations that are not divisible (Rogers, 1995). It is an essential feature of innovation because it provides a means for prospective adopters to reduce the uncertainty they feel toward an unfamiliar technology or product (Weiss & Dale, 1998). Moreover, Jung Moon (2020); Putzer & Park (2012) affirm that trialability is a strong determinant of attitude towards a behaviour. Observability indicates the degree to which the innovation provides tangible results. Observability is challenging to establish in a market already dominated by an established technology, yet it is a crucial motivator for adoption (Rogers, 2003). Observability is a strong determinant of attitude toward a smartphone (Putzer & Park, 2012) and influences consumer intentions for green places (Chou *et al*., 2012). On the contrary, Jung Moon (2020) argued that the observability of electric vehicles does not affect consumer attitudes significantly.

DOI has proven a successful theory for investigating technology adoption in communication, agriculture, public health, criminal justice, and social work, among others (Ismail, 2016). However, despite the theory's greatness in adoption research, it has received criticism. According to Hayden (2014), DOI does not consider personal resources to embrace new technology, and it is more appropriate to embrace behaviours instead of cessation. Although these criticisms are vital, this study could not overlook this theory; however, to supplement it, the theory of planned behaviour that embraces cessation of behaviour was also taken into consideration.

2.3.4 Theory of Planned Behaviour (TPB) and adoption

TPB was propounded by (Ajzen, Icek, 1985,1991) as an extension of the Theory of Reasoned Action (TRA), which was also first developed by Martin Fishbein in the late 1960s. TRA posits that the main predictor of voluntary behaviour is one's intention (Glanz *et al.*, 2015). This intention is recognised as behavioural intention and results from attitudes and subjective norms (Colman, 2015; Doswell *et al*., 2011). Accordingly, attitudes are the personal feelings about a specific behaviour, determined by the strength of beliefs concerning the outcomes of the performed behaviour and the evaluation of the potential outcomes (Fishbein & Ajzen, 1975; Ajzen & Albarracín, 2007). Fishbein (1967) continued to ascertain that these attitudes can be positive, negative or neutral.

On the other hand, subjective norms refer to how perceptions of relevant groups such as family members, friends, and peers may influence one's performance in the behaviour. Ajzen also defines subjective norms as the perceived social pressure to perform or ignore the behaviour (Ajzen & Albarracín, 2007). As its focus is on voluntary behaviour where a person chooses whether to perform or ignore an action, the application of TRA is limited and considered deficient primarily where obstructions to performing a behaviour exist.

Given the limitation, TPB was developed by adding a third determinant, perceived behavioural control (PBC), to the first two determinants to account for behaviour hurdles to acting (Cook *et al*., 2002). PBC is the degree of difficulty in performing a behaviour, notably, the extent to which an individual believes that he or she can perform a given behaviour (Daxini *et al*., 2018). As such, perceived behavioural control is behaviour-specific and encompasses the perception of the person's ability to effectuate the behaviour. Fishbein & Ajzen. (1980) continues to assert that perception varies by environmental circumstances and the behaviour involved. Therefore, Cook et al. (2002) conclude that perceived behaviour control mediates other predictors of intention since a person may have a positive attitude towards a behaviour but may abandon it when faced with an obstacle. TPB suggests that people are far more likely to intend to perform certain behaviours when they believe that they can enact them adequately as influenced by their attitude, subjective norms and perceived behaviour control (Borges *et al*., 2014; Oparinde *et al*., 2017).

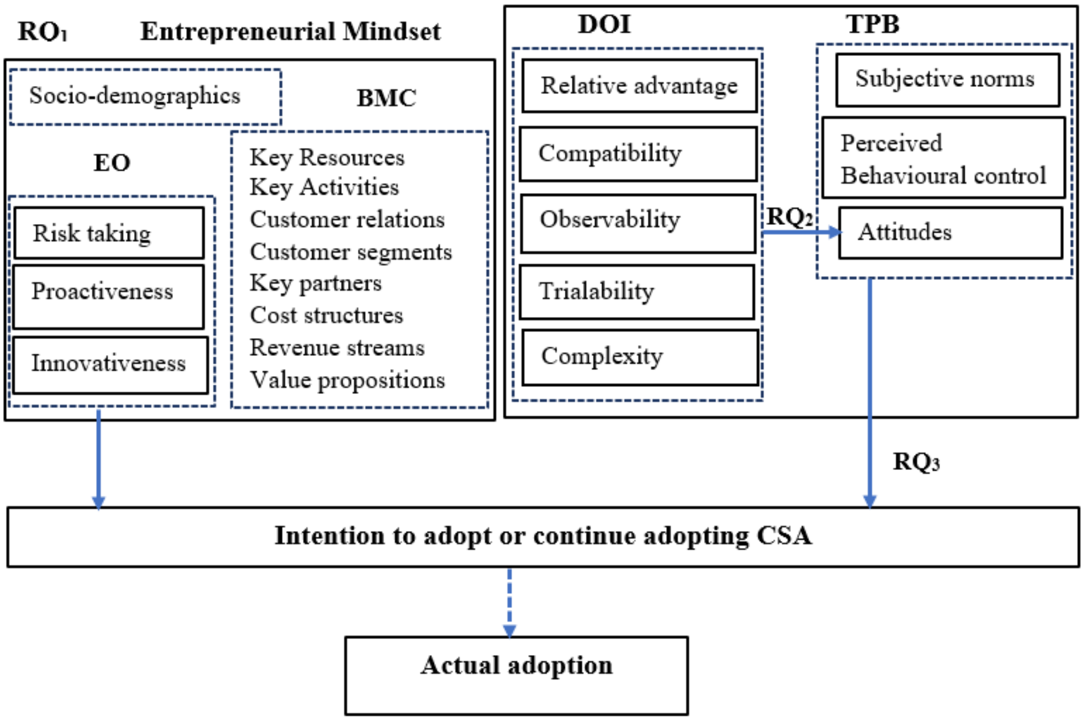
TPB has been utilized in a wide range of behavioural science disciplines and has been used empirically in a variety of situations to explain and understand behaviour, for instance; student decisions regarding their career path (Ingram *et al*., 2000), motivations of students regarding choice of courses (Pineda, 2009), and explaining sexual behaviour (Doswell *et al.*, 2011). It has also been vital in studying farmers' adoption of sustainable agricultural practices (Menozzi *et al*., 2015), predicting consumers' green hotel visit intention (Verma & Chandra, 2018), and adopting preventive behaviour against COVID-19 (Bronfman *et al*., 2021) among others. Regardless of the context, most TPB studies have concentrated on consumers and students, and only a few focused on farmers. It is, therefore, crucial to analyze farmers' behavioural intention to understand differences explaining why CSA technologies may/may not end up in the fields of coffee farmers.

Many empirical studies have investigated the determinants of behavioural intention. Although all the three elements of the TPB significantly predict behavioural intention, the attitude is the strongest and most consistent predictor of behavioural intention (Chen, 2017; Prati *et al*., 2012). While several studies noted a significant and positive influence of attitude on intention (Cristea & Gheorghiu, 2016; Dalila *et al*., 2020; Hasan & Suciarto, 2020; Mead & Irish, 2021; Shan *et al*., 2020), a few found it insignificant (Chen, 2017; Noor *et al*., 2020; Talsma *et al.*, 2013). Kim *et al.* (2013) noted that attitude is a mediator in the relationship between subjective norm and behavioural intention. Subjective norms positively affect behavioural intention (Dalila *et al*., 2020; Ibrahim & Arshad, 2017; Lee, 2011; Mead & Irish, 2021). Nevertheless, Cera & Furxhiu (2017; Hasan & Suciarto (2020; Smith (2015) found a significant negative link between Subjective norms and behavioural intention.

Different studies have established mixed findings regarding the association between perceived behavioural control and behavioural intention. Notably, perceived behavioural control positively and significantly affect behavioural intention (Cera & Furxhiu, 2017; Cristea & Gheorghiu, 2016; Hasan & Suciarto, 2020; Lee, 2011; Noor *et al*., 2020; Shan *et al*., 2020). Meanwhile, an insignificant relation exists between perceived behavioural control and intentions (Ibrahim & Arshad, 2017; Smith, 2015) and a negative effect (Anggraini, 2016; Mead & Irish, 2021; Prati *et al*., 2012). Prati *et al*. (2012) observed that in all the studies, the measurement of the constructs was by different instruments, hence the variability of the findings. Nonetheless, TPB variables, especially attitudes, play a crucial role in forming behaviour. This study proposes a relationship between TPB variables and the farmers' intention to adopt CSA technologies in the greater Luweero region, Uganda.

2.3.5 Conceptualizing Diffusion of Innovation theory and the Theory of Planned Behaviour

The DOI and TPB have been applied jointly in studies of information systems research (Weigel *et al*., 2014), adoption of electric vehicles (Jung Moon, 2020), and fertilizer behaviours in landscape management (Warner *et al*., 2021); however, studies relating to farmers are limited. Meanwhile, both theories have been used separately in a wide range of empirical studies relating to agriculture, information technology, consumer behaviour, trade standards and nutrition (Kasperavičiūtė-Černiauskienė & Serafinas, 2018; Mairura *et al*., 2016; Oparinde *et al.*, 2017; Verma & Chandra, 2018). The general use of DOI in agricultural innovation research has proven that the perceived characteristics of the technology influence the adoption–decision process. However, according to Glover *et al*. (2019), DOI considers technology adoption largely individually influenced, relatively simple, and a linear progression from old inferior methods to new ones. Secondly, DOI is more appropriate in embracing behaviours instead of cessation of behaviours (Hayden, 2014). As Weigel *et al*. (2014) pointed out, introducing variables that affect the decision makers' intention and behaviour gives additional explanatory power regarding the adoption decision. Like Warner *et al*. (2021), all TPB variables predict behavioural intent, but a combined model with DOI's technology attributes provides a solid base for analysing the adoption-decision process as both models are focused on the decision maker's perception. As such, the study believes that the attributes of TPB, when combined with the technological qualities described in DOI, provide extra predictive power for the intention to adopt CSA. Recent studies have leveraged the TPB to examine the influence of behavioural variables on the intention to adopt CSA (Price & Leviston, 2014; Van Hulst & Posthumus, 2016; Zeweld *et al*., 2017), thus justifying the exploration of technological characteristics that influence attitudes toward adoption of CSA. Moreover, their combined use is vital to capture the complex social and technical components of CSA adoption. Yet, the TPB and DOI have rarely been considered together regarding farmers' intention to adopt CSA. This study adds to the limited literature on understanding the influence of technological and behavioural factors on farmers' attitudes and adoption, by empirically examining the Intention to adopt or continue adopting CSA technology by coffee farmers.



Abbreviations: RQ1-3, Research Question (1-3). EO, Entrepreneurial Orientation. BMC, Business Model Canvas. TPB, Theory of Planned Behaviour. CSA, Climate-Smart Agriculture. → Association between variables

Figure 2- 1: Conceptual framework (Own compilation)

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter discusses the research philosophy and technique used to investigate the relationship between technology characteristics, behaviour factors, farmers' entrepreneurial mindset, and their intentions to adopt CSA in greater Luwero, Uganda. It also includes a description of the study area, target population, a sampling design, a data collection instrument, validity and reliability, a data collection technique, data analysis, and the results presentation plan, as well as ethical considerations.

3.2 Description of the study area

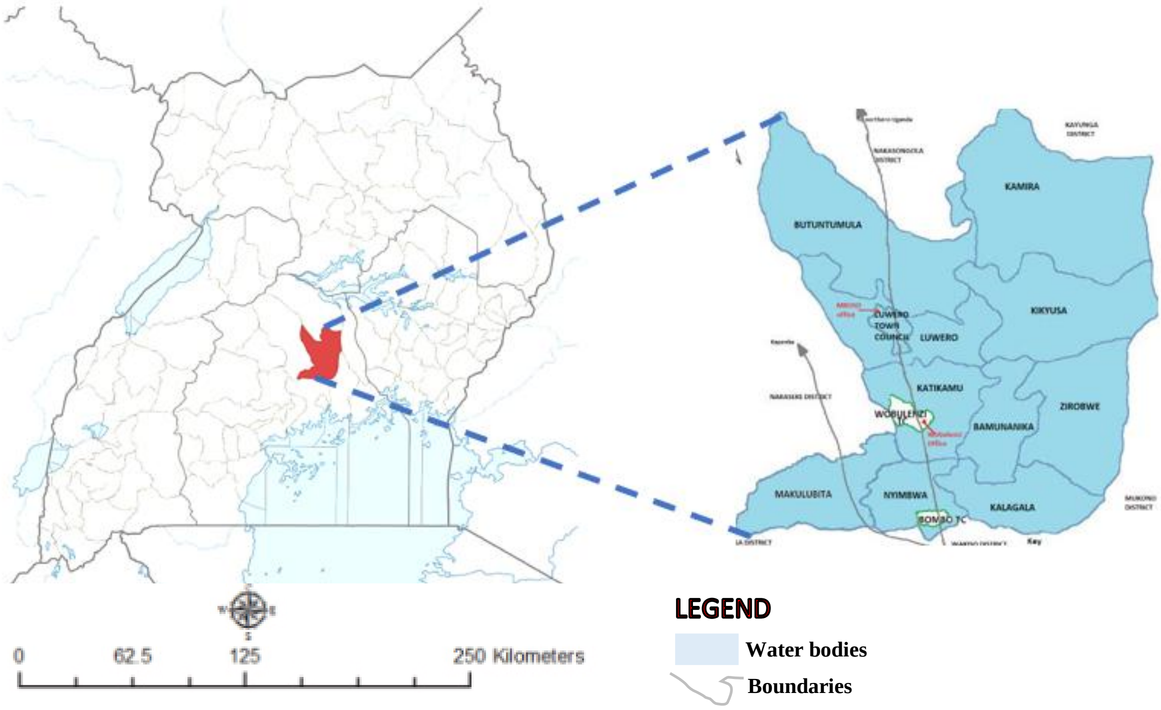


Figure 3- 1: Map of Uganda showing the study area

The study site was Greater Luwero, located in Central Uganda, 75 km by road, North of Kampala, Uganda's capital, at latitude 0^{0}50'57.01" N and longitude of 32^{0}28'23.02" E. The total area of the Luwero district is approximately 2577.49 Sq. Kilometres. On its borders, there are Mukono and Wakiso districts in the South, Nakaseke in the West, Nakasongola in the North and Kayunga district in the East. Regarding altitude, the Luwero district ranges between 1,219 and 1,524 meters above sea level. The landscape generally comprises an elevated and dissected plateau with a series of flat-topped hills and intervening valleys onto which farmers plant the crops. With a mean annual rainfall of 1,300mm, precipitation is evenly spread throughout the year.

The mean annual maximum temperature falls between 27.5^{o}C and30^{o}C, whereas the mean annual minimum temperature is between 15^{o} C and 17.5^{o}C. In this generally rural area, farming is the primary occupation, especially for women. This region was chosen because of the dominance of coffee production, and it is one of the regions where CSA projects were first launched in Uganda (USAID & CIAT, 2017). Coffee is a strategic crop in Uganda, making it attractive for the fast-growing espresso industry. Coffee is a crucial cash crop and a major source of income for many smallholder coffee growers, but their low yields limit their earnings, affecting their livelihoods. Coffee farmers in greater Luweero are approximately 55,126, which is almost 10% of the total population of coffee farmers in the Central region of Uganda, mainly producing Robusta coffee (UBOS, 2016). Robusta coffee is native to Uganda, and the two major types grown in Luweero are; Nganda and Erecta. Nganda is a smaller, less resistant species. It does not produce fruit throughout the year, except at the end of the two rainy seasons, October and June. Plants of the Erecta can produce for decades and reach a height of ten meters (Wang *et al.*, 2015). This coffee variety is resistant to all the major coffee diseases. Despite the government's push to replace traditional types with more productive commercial varieties, many coffee farmers have decided to maintain the Nganda coffee (Bunn *et al.*, 2019). Coffee is primarily grown on mixed farms, mainly rainfed under a low input system and intercropped with bananas, soya, and beans to ensure households' food security (FAO, 2020a). It is also grown among shade trees that act as wind breakers in the garden. Farmers in Luweero use family labour on their farms which is cheaper in the long run and readily available.

3.2 Target Population

The study targeted coffee farmers in the Luwero district. According to the International Coffee Report Council (ICO) (2019), the district has 55,126 smallholder coffee farmers. There are different categories of coffee farmers, including those practising CSA and those that have not yet embraced the CSA technologies. For this study, the unit of observation is the coffee farm, and the unit of analysis is the coffee farmer who is not yet implementing all the recommended CSA technologies for proper coffee management as suggested under the Climate Smart Investment Pathway (Stepwise) (Jassogne *et al.*, 2017). The target population spreads across 13 sub-counties that comprise the Luwero district.

3.3 Sampling

The study used a cross-sectional research approach to collect data among coffee farming households. The choice of respondents followed a multi-stage sampling technique since it is cost and time-effective when collecting data from a population that is geographically dispersed. The technique followed three steps of sampling as described below. Firstly, purposive sampling to choose the Luwero region because CSA technologies were launched and promoted in this area. The area has thirteen sub-counties. Secondly, two sub-counties were selected from the district following a simple random sampling approach to ensure an equal chance of being selected. Ultimately, choosing one coffee producer organisations randomly from each sub-county thus Nakikaaka and Kyalugondo coffee farmers' association. All smallholder coffee farmers who were not implementing all the recommended CSA technologies under the Climate Smart Investment Pathway were selected with their address/contact information using a credible database obtained from the coffee producer organisations. 230 coffee farmers made up the study's sample size, which is appropriate for most research (Sekaran & Bougie, 2003). Moreover, it is sufficient to conduct structural equation modelling analysis as the minimum required sample to run a structural equation model is 100 respondents (Hair *et al*., 2010).

3.4 Ethical consideration

We obtained a written ethical clearance to conduct the study from the International Institute of Tropical Agriculture (IITA) and an ethical clearance certificate obtained from its Internal Review Board (IITA IRB) in accordance with the IITA Research Ethics Policy as data was collected from its project sites and areas of operation. The research assistants introduced themselves to the smallholder coffee farmers and explained the study's aim and significance. A good relationship was built with the respondents at the beginning to gain their hope and confidence.

3.5 Data collection

We collected data from March to April 2022 using a digitized structured questionnaire to guide face-to-face interviews in a household survey. The survey execution proceeded with the help of 10 research assistants after training them on ethical issues, data significance, data quality and data collection methods. Each research assistant quickly clarified the study's goal to the respondents at the start of each interview, assuring them of the anonymity of their responses and the importance of their opinions. Only farmers growing coffee for three years or longer were considered for the study, as coffee matures and gives optimal yield at three years. Hence, the selected farmers were able to provide reliable information about the farm management and yield of their coffee. Only coffee farmers aged 18 years and above were allowed to participate in the study, as this is Uganda's adulthood age. After every interview, the completed questionnaires were immediately uploaded to the Open Data Kit server by the research assistants.

3.5.1 Instrument for data collection

There were six sections to the structured questionnaire: Section A included structured questions about socioeconomic and farm characteristics; Sections B and C included structured questions about DOI and TPB components; Section D had questions about coffee farmers' intention to adopt CSA practices, which is the study's dependent variable and Section E and F contained structured questions about Entrepreneurial Orientation and BMC compliance for categorizing farmers. The administering of the digitized structured questionnaire was by use of tablets and mobile phones. The farmers received general information describing Climate-Smart agriculture based on the three principles stated by FAO (2013), and this was the basis for assessing farmers' perceptions of climate-smart agriculture technologies.

3.5.2 The theory of Planned Behaviour, Diffusion of Innovation Theory, Entrepreneurial Orientation Theory and Business Model Canvas instrument

The assessment of the DOI, TPB and EO constructs used a 7-point Likert scale. For each statement under a specified construct, farmers ranked their opinions from 1 (strongly disagree) to 7 (Strongly agree). BMC element compliance assessment used a 5-point Likert scale where 1 corresponds to disagree strongly, and 5 strongly agree. The development of the DOI part of the CSA adoption scale used items falling into specific attributes for Rogers (1962) subscales, namely "Relative Advantages", "Compatibility", "Trialability", "Complexity", and "Observability" based on a previous study by Moore & Benbasat (1991) measuring the perceptions of adopting an information technology innovation. To create the study constructs, Moore & Benbasat (1991) 15 items were reworded to refer to the CSA innovation rather than information technology (Appendix 1). For the case of relative advantage, we measured it using five items. We used three items for the others (compatibility, complexity, trialability and Observability).

Similarly, 7-point Likert scales were used to assess adoption intentions, attitudes, subjective norms, and perceived behavioural control. All constructs were adapted from previous research and created using the guidelines outlined by Ajzen (2006). However, we altered the items to meet the CSA adoption context in Ugandan coffee farming (Appendix 1). Each of the constructs (attitude, perceived behavioural, subjective norms and adoption intention) was measured using three items. Finally, coffee farmers assessed their intentions to adopt CSA by responding to the statements to indicate their level of agreement or disagreement. For EO, we followed Miller's (1983) three dimensions conceptualization. Miller/Covin and Slevin's (1989) nine-item scale guided the creation of the measurement items. They were, however, slightly modified to meet the Ugandan setting of coffee production (Appendix 1). The items consisted of pairs of opposite statements divided by a seven-point scale. Lumpkin & Dess (2001) contended that the Covin and Slevin question, which asked whether a company prefers to 'undo competitors,' assesses competitive aggression rather than proactiveness, so the data collection instrument did not include this question. The overall farmers' entrepreneurial orientation is a combination of individual assessment of innovativeness, risk-taking, and proactiveness constructs. Results presentation for all the items appears in chapter four (Table 4.5), which displays the factor and factor scores found in this study.

In February 2022, a pretest was conducted in the Nakaseke district to screen the questionnaire for length, content, and comprehension. Cronbach alpha coefficients were computed for the individual item component variables (using scale purification techniques as appropriate) and the overall constructs to test the construct's internal reliability. However, in the latter case, only the individual scores of the variable scales were used as input to compute the overall construct's alpha coefficient.

Table 3. 1: Operational definition of the DOI, TPB and EO constructs

Construct	Operational definition
Relative advantage	The extent to which CSA technologies are superior to other technology.

Compatibility	The degree to which CSA technologies are perceived as consistent with coffee farmers' existing values, experiences, and needs.
Complexity	The degree to which CSA technologies are perceived as relatively difficult to understand and implement.
Trialability	The extent to which coffee farmers can quickly test CSA technologies before being applied.
Observability	The degree to which CSA technologies provide tangible results.
Attitude	The coffee farmer's positive or negative evaluation of using CSA technologies.
Subjective norm	The perceived social pressure to use or not to use CSA technologies.
Perceived behavioural control	The coffee farmer's confidence in his ability to implement CSA technologies.
Risk-taking	The coffee farmer's willingness to undertake risky resource commitments to attain specific goals.
Innovativeness	The ability of a coffee farmer to generate new ideas, uncover new market opportunities, and engage in creative processes.
Proactiveness	The ability of a coffee farmer to act in advance of a future situation, tending to initiate change rather than reacting to change

3.6 Data analysis

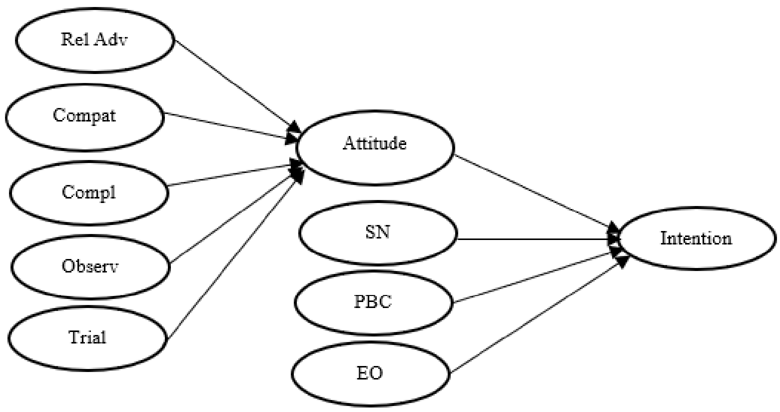
The collected data was exported from the Open Data Kit server to the Statistical Package for Social Sciences (SPSS) version 27 and edited to ensure perfectness, absoluteness, and coherence. Multiple-choice variables like the adoption of individual CSA technologies were turned into dummy variables, after which composite variables were developed from the dummy variables of adoption of individual technologies but also for entrepreneurial orientation scale variables. Furthermore, R software version 4.0.5 was also used to analyze structural modelling and confirmatory factor analysis. An explanatory research approach was used to verify the relationship between study variables and explain the reasons for an observed event.

3.6.1. Analysis of Socio-demographic characteristics, integrated DOI, TPB and EO constructs

The socio-demographic and farm characteristics were explicitly analysed using descriptive statistics (frequencies, percentages, mean and standard deviations) and graphing techniques to abridge the nature of these characteristics. Confirmatory factor analysis (CFA) was used to verify the observed variables' factor structure for EO. The hypothesis that there was a link between observable variables and their underlying latent components was tested using CFA with the package 'lavaan', which was constructed under R version 4.0.5. The factor loadings were extracted from the CFA model summary and used to assess the reliability and validity of the CFA model. The Chi-square test, absolute fit indices, incremental fit indices, and parsimonious fit indices were also used to analyze the model's goodness of fit (GOF). According to Hair *et al*. (2010), the rule of thumb for GOF is to use the Chi-square test and at least one index from each of the other groups.

The verified EO constructs were then used as grouping variables to distinguish a relatively homogeneous group of coffee farmers by hierarchical cluster analysis. In the cluster analysis, the three EO constructs composite variables based on mean scores of the observable variables of Risk-taking, Innovativeness, and Proactiveness were entered as cluster variables. Ward's cluster method was utilized based on squared Euclidean distances. Cluster membership was examined utilizing profiling variables linked to socio-demographics and farm characteristics, as well as compliance to business model canvas elements, after selecting the most acceptable cluster solution (i.e., four clusters). The analysis of variance (ANOVA) test was used to compare means with profile variables to discover significant variations between the four clusters.

Because we had two dependent variables, we used structural equation modelling (SEM) with the package 'lavaan', which was constructed under R version 4.0.5 for the DOI, TPB, and EO constructs. SEM is a multivariate technique that predicts a number of interconnected dependent connections simultaneously to see how well it matches the data. SEM consisted of two models: (1) a measurement model, which generated the latent constructs as a function of the observed variables; and (2) a structural model (figure 3.2) that quantified the relationships between the latent constructs (Hair *et al*., 2010). The model's goodness of feet was assessed using the Chi-square ratio (CMIN/DF), Tucker–Lewis index (TLI), Comparative Fit Index (CFI), and Standardized Root Mean Squared Residual (SRMR) and the Root Mean Square Error of Approximation (RMSEA). Other researchers have also used SEM in their studies, for example, Buyinza *et al*. (2020) and Dang *et al*. (2018).



Abbreviation: Rel Adv, Relative advantage. Compat, Compatibility. Compl, Complexity. Trial, Trialability. Observ, Observability. ATB, Attitude to behaviour, SN, Subjective norms. EO, Entrepreneurial Orientation. PBC, Perceived behavioral control.

Figure 3- 2: Structural model

3.6.2 Reliability and validity tests

These tests allowed the researcher to determine whether the data was suitable for structural equation modelling and, in particular, confirmatory factor analysis. Composite reliability (CR), internal consistency (Cronbach's Alpha), and average variance explained were used to assess the reliability of each concept in the measurement model (AVE). CR and Cronbach's Alpha values vary from 0 to 1, but values of 0.60 and higher are acceptable, while values lower than 0.6 are rejected (Hair *et al*., 2010). Similarly, AVE values of 0.5 and above are accepted, and values below are rejected (Ahmad *et al*., 2016). The formula to calculate the value of

Composite Reliability (CR) and Average Variance Extracted (AVE) is shown in the table below.

Table 3. 2: Reliability formulae

Measure	Formula
CR	$(\Sigma\kappa)^2 / [(\Sigma\kappa)^2 + (\Sigma 1 - \kappa^2)]$
AVE	$\Sigma\ \kappa^2 / n$

K = factor loading of every item, n = number of items in a model. Source: (Ahmad *et al*., 2016)

Model validity was assessed using the goodness of fit measures (GOF), model significance, and the direction and size of structural parameter estimates. These are the specified measures of model validity by Hair *et al*. (2010). Discriminant validity was also assessed using the correlation matrix of latent exogenous constructs. The accepted values are below 0.85, and the square root of AVE for the construct must be higher than the correlation between respective constructs (Ahmad *et al*., 2016).

CHAPTER FOUR: RESULTS PRESENTATION

4.0 Introduction

This chapter provides an overview of the socio-demographic and farmer characteristics, descriptive statistics, reliability and validity tests for the measurement model, and the structural equation model parameter estimation.

4.1 Socio-demographic and farm characteristics

According to Table 4.1 below, a total of 230 coffee farmers of both genders participated in the survey. It is not surprising that most coffee farmers were male (67.8%), and females accounted for 32.2 %. Furthermore, most smallholder coffee farmers (76.5%) were married, 14.3% were widowed, 6.5 % were single (never married), and 2.6 % were divorced/separated. Except for 35 respondents (15.2%) who were not the heads of their households, most respondents (84.8%) doubled as their household heads. The mean age of the coffee farmers was 52.62 ± 14.831 years, and on average, the number of years they spent in formal education was 6.40 ± 4.811. The average size of a coffee farm was 1.46 ± 1.831 hectares, of which 0.14 ± 0.556 accounted for hired land. Regarding household earnings, the average monthly household income was USD 90.8 ± 206.358, on which an average household size of 7.47 ± 4.523 members depends. Concerning the farming experience, respondents reported an average of 29.45 ± 15.035 years spent practising agriculture, of which 23.80 ± 14.438 years were in coffee farming.

Table 4. 1: Farmers' sociodemographic and farm characteristics

Response	F	%
Gender		
Female	74	32.2
Male	156	67.8
Total	**230**	**100.0**
Marital Status		
Divorced/Separated	6	2.6
Married	176	76.5
Single	15	6.5
Widow(er)	33	14.3
Total	**230**	**100.0**
Household Head		
Yes	195	84.8
No	35	15.2
Total	230	100
	Mean	**St.dev**
Age	52.62	14.831
Years of formal education (years)	6.40	4.811
Household Size	7.47	4.523
Household Monthly income (USD)	90.8	206.358
Farm size (ha)	1.46	1.831
Hired land (ha)	0.14	0.556
Years spent in agriculture (years)	29.45	15.035
Years spent in coffee farming (years)	23.80	14.438

Source: Survey data.

4.2 Information about CSA and business training

Coffee farmers were asked to report their access to CSA information and attendance at a business skills training using yes/no questions. The bar graph below shows that the majority of the respondents (67.80%) had never accessed CSA information as opposed to the 32.20% who reported to have accessed CSA information. Similarly, a bulk of respondents (54.80%) had never attended a business skills training, even though 45.20% reported that they had attended such training, as shown in figure 4.1 below.

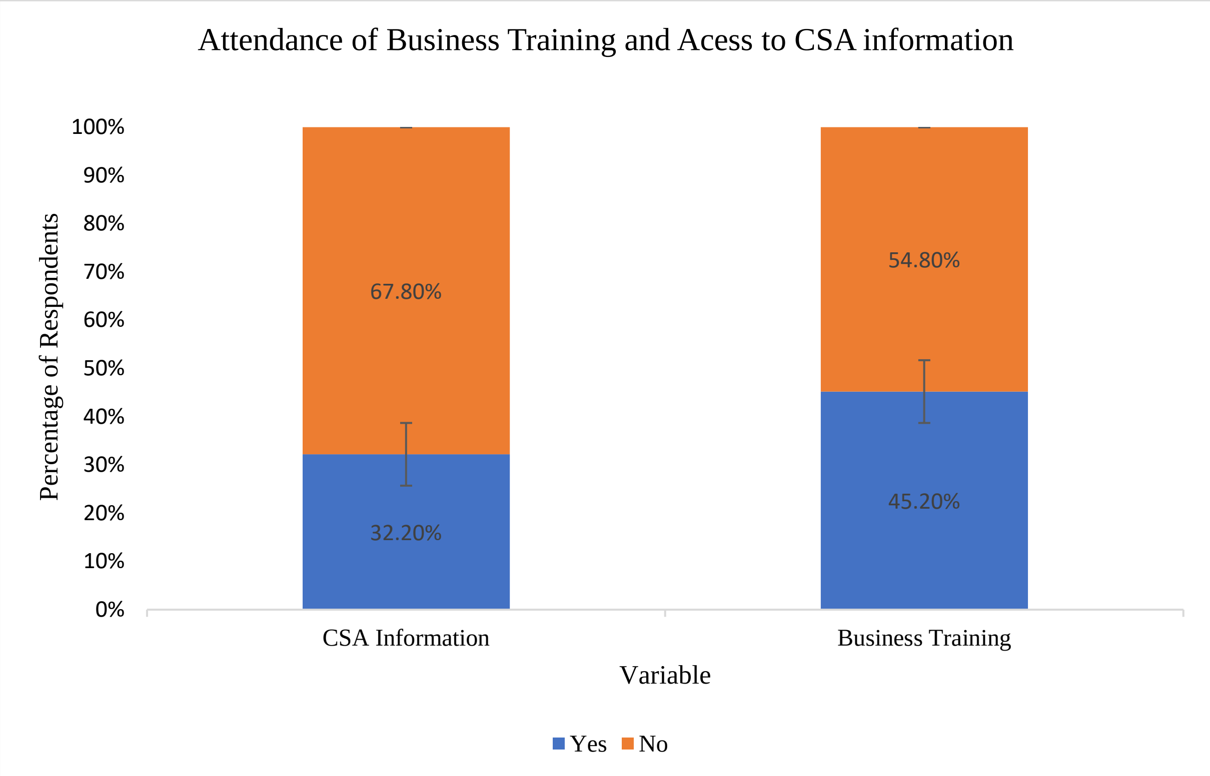


Figure 4- 1:Coffee farmers' response on their attendance of business training and access to information about Climate-Smart Agriculture technologies. The percentages are based on the total number of respondents, N= 230

4.3 Farmers' Entrepreneurial Mindset

4.3.1 Confirmatory factor analysis for farmers' Entrepreneurial Orientation (EO)

Table 4.2 presents the results of factor loadings, reliability, internal consistency and average variance extracted for the EO constructs. Risk1, Risk2 and Risk3 denote the observed variables of the Risk-taking construct. Similarly, those of Innovativeness and Proactiveness are denoted by Inno1, Inno2 and Inno3 and Proact1 and Proact2, respectively. The value of factor loading for all items is greater than 0.6. Other than that, the value of Cronbach Alpha for all constructs is greater than 0.60 and the Average Variance Extracted (AVE) for each construct is greater than 0.5

Table 4. 2: Reliability and validity test for the EO measurement model

Construct	Item	Factor loading	*P*-value	Construct Reliability (CR)	Average Variance Extracted (AVE)	Cronbach's Alpha
Risk taking	Risk1	0.883	**0**	0.908	0.767	0.908
	Risk2	0.908	**0**			
	Risk3	0.834	**0**			
Innovativeness	Inno1	0.693	**0**	0.856	0.667	0.836
	Inno2	0.799	**0**			
	Inno3	0.753	**0**			
Proactiveness	Proact1	0.899	**0**	0.880	0.766	0.868
	Proact2	0.860	**0**			

Source: Survey data.

Table 4. 3: EO Model fit indices

Statistic	Value
CMIN/DF	1.989
TLI	0.984
CFI	0.994
RMSEA	0.066
SRMR	0.018
χ^2,**df**,p	19.891, 10, 0.030

Source: Survey data.

Based on the CFA calculations, a fitting model was calculated with a probability value of 0.030, a CMIN/DF score of 1.989, a CFI score of 0.994, an RMSEA score of 0.066 and an SRMR score of 0.018 (Table 4.3).

4.3.2 Segmentation of farmers according to their Entrepreneurial Orientation

Hierarchical cluster analysis was used to classify coffee producers based on three characteristics that measured their entrepreneurial orientation: Risk-taking, Innovativeness, and Proactiveness and yielded four clusters. Non-enterprising farmers (31 farmers and make up 13.5 % of the sample), indecisive farmers (94 and makeup 40.9 %), conservative farmers (58 and makeup 25.2 %), and enterprising farmers (47 and make up 20.4 %). Table 4.4 reveals statistically significant differences in the means of cluster variables between the clusters, with enterprising farmers having the highest risk-taking, innovativeness, and proactiveness and non-enterprising farmers having the lowest means for all constructs. The results of cluster profiling using socio-demographic variables and attention to business model canvas features are also summarized (Table 4.4). The clusters differed considerably on eight variables: respondent age, years of farming experience, total land, monthly household income, customer groups, value propositions, revenue streams, and cost structures.

Regarding the business model canvas, enterprising farmers pay close attention to customer segments and value propositions while paying only moderate attention to revenue streams and cost structures. Though not statistically significant, it is surprising that non-enterprising farmers pay close attention to key partners, communication channels and revenue streams. The rest of the business model elements also did not present significant differences. In terms of socio-demographics, as expected, enterprising farmers had more farmland and the highest household monthly income, which were statistically significant. Finally, the differences between the years of education, coffee farming, and the size of hired land were not statistically significant.

Table 4. 4: Group-specific descriptive ANOVA mean and (sd) Overall, then groups

		Entrepreneurial Mindset				*P*-value
		Non-enterprising (13.5%) Mean(SD)	Indecisive (40.9%) Mean(SD)	Conservative (25.2%) Mean(SD)	Enterprising (20.4%) Mean(SD)	
Socio-demographics	Age (yrs)	42.23(14.387)	54.72(14.648)	55.19 (14.040)	52.11(14.324)	<.001*
	Years of education (yrs)	8.16(9.434)	6.56(3.555)	5.52(4.005)	6.00(2.912)	.087
	Years of Agriculture (yrs)	20.32(14.650)	31.05(14.608)	32.24(15.473)	28.81(13.623)	.002*
	Years of coffee farming (yrs)	18.06(15.526)	25.76(14.162)	24.47(14.515)	22.83(13.512)	.072
	Total land (ha)	1.29(0.949)	1.34(1.172)	1.22(0.765)	2.13(3.468)	.046*
	Hired land (ha)	0.13(0.248)	0.10(0.344)	0.12(0.256)	0.26(1.076)	.429
	Monthly income (USD)	51.67(52.841)	59.66(149.064)	88.56(208.052)	164.49(294.811)	.018*
Entrepreneurial Orientation	Risk taking[a]	2.60(0.389)	3.20(0.401)	4.76(0.537)	5.59(0.369)	<.001*
	Innovativeness[a]	3.05(0.431)	3.45(0.434)	4.45(0.639)	5.67(0.489)	<.001*
	Proactiveness[a]	2.42(0.291)	3.18(0.290)	4.78(0.430)	5.74(0.476)	<.001*
Business Modal Canvas attention	Customer segments	2.80(0.601)	3.00(0.9274)	3.33(1.015)	3.49(1.040)	.003*
	Value propositions	2.87(0.619)	2.98(0.604)	3.48(0.662)	3.66(0.939)	.012*
	Communication channels	4.00(0.516)	3.87(0.765)	3.78(0.773)	3.62(0.848)	.128
	Customer relations	2.03(0.604)	2.14(0.784)	2.28(0.790)	2.38(0.873)	.169
	Key resources	2.35(0.985)	2.78(1.049)	2.98(1.084)	2.83(1.129)	.071
	Revenue streams	3.23(1.146)	3.15(1.107)	2.67(1.082)	2.91(1.158)	.045*
	Key activities	3.06(1.169)	3.30(0.993)	3.03 (1.123)	3.48(0.996)	.164
	Key partners	3.29(1.006)	3.18(1.100)	2.81(1.047)	2.96(1.141)	.107
	Cost structures	2.61(1.202)	2.81(1.100)	3.16(0.931)	2.87(1.115)	.045*

N=230, * Statistical significant at 5%, [a] clustering variable, Abbreviation: SD, standard deviation.
Source: Survey data.

4.3.3 Evaluation of Perceived Compliance to Business Model Canvas (BMC) elements

Figure 4.2 depicts the coffee farmers' evaluation of their conformity to BMC elements based on two indicators (information and attention) of the nine BMC elements. Regardless of the compliance indicator, the customer segments and value proposition have the highest mean values for the two indicators. Except for communication channels, key partners and cost structures, smallholder coffee farmers have more information about BMC elements compared to the attention they pay to them. It is surprising how smallholder coffee farmers have little information (2.50) and pay minimal attention (3.18) to the cost structures of their coffee business and the revenue streams at 2.76 and 3.02 for information and attention, respectively.

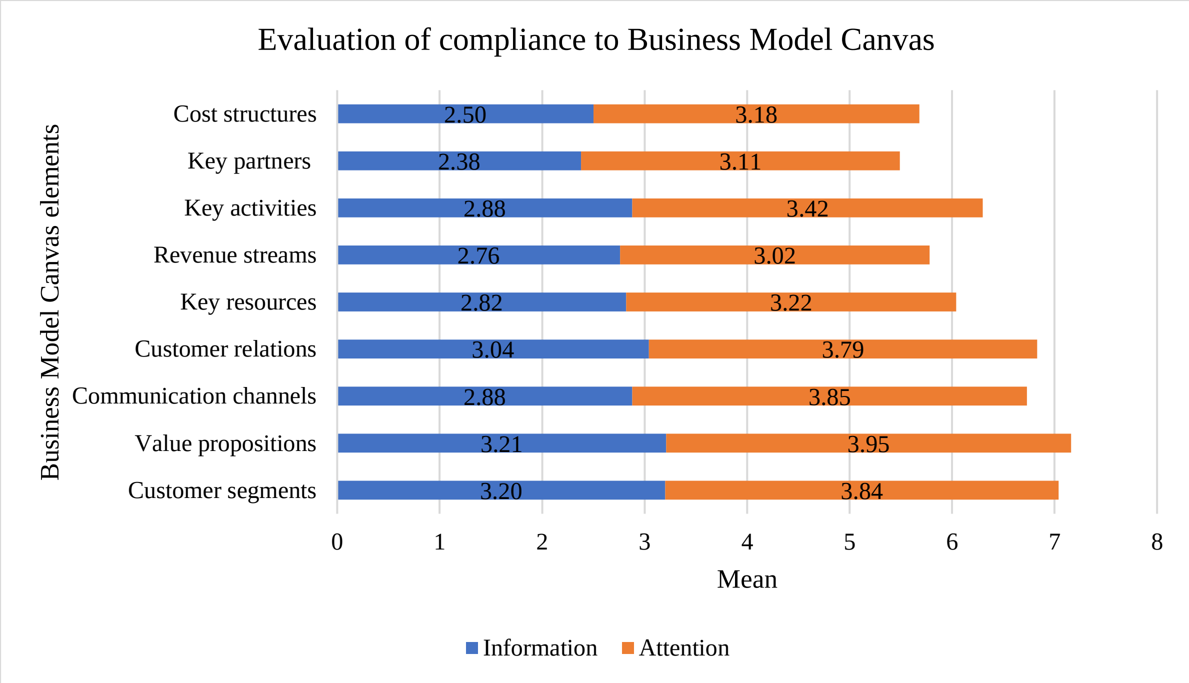


Figure 4- 2:Evaluation of farmers' compliance to business model canvas elements in terms of the information and attention they pay per element. Mean values were based on a 1–4 scale for information about the BMC elements and a 1-5 scale for the attention paid to each BMC element.

4.4 Reliability and Validity of the Measurement Model

Table 4.5 shows results from the three dimensions used to gauge the model's reliability; the internal consistence of the scale's items was based on Cronbach's Alpha. Composite reliability was based on the factor loading same as the Average Variance Extracted (AVE).

Table 4. 5: Reliability test for the measurement Model

Construct	Item	Factor loading	Composite Reliability (CR)	Average Variance Extracted (AVE)	Cronbach's Alpha
Relative Advantage	Rel1	0.879	0.951	0.796	0.951
	Rel2	0.905			
	Rel3	0.909			
	Rel4	0.915			
	Rel5	0.850			
Compatibility	Compat1	0.939	0.962	0.894	0.962
	Compat2	0.962			
	Compat3	0.936			
Complexity	Compl1	0.781	0.904	0.758	0.901
	Compl2	0.904			
	Compl3	0.921			
Trialability	Trial1	0.726	0.843	0.642	0.834
	Trial2	0.868			
	Trial3	0.804			
Observability	Observ1	0.925	0.934	0.824	0.931
	Observ2	0.949			

	Observ3	0.846			
Perceived Behavioural Control	Per1	0.900	0.921	0.795	0.917
	Per2	0.900			
	Per3	0.875			
Subjective Norms	Subj1	0.897	0.956	0.880	0.956
	Subj2	0.975			
	Subj3	0.940			
Attitude	Att1	0.913	0.911	0.775	0.908
	Att2	0.922			
	Att3	0.800			
Entrepreneurial Orientation	Risk1	0.863	0.952	0.740	0.952
	Risk2	0.861			
	Risk3	0.897			
	Inno2	0.828			
	Inno3	0.778			
	Proact2	0.893			
	Proact1	0.896			
Intention to adopt	adop1	0.930	0.945	0.852	0.942
	adop2	0.950			
	adop3	0.887			

Source: Survey data.

Cronbach Alpha values for all constructs in the measuring model were higher than the 0.5 thresholds and exceeded the 0.6 level (Hilton *et al*., 2014). As a result, Internal Reliability met the needed level; however, compatibility was the most internally reliable, while trialability was the least. All values of CR are larger than 0.6, so Composite Reliability was achieved (Ahmad *et al*., 2016). Finally, the measurement model achieved the Average Variance Extracted at a required level of more than 0.5.

Table 4. 6: Discriminant validity

Construct	Rel Adv	Compat	Compl	Trial	Observ	ATB	SN	EO	PBC	INT
Rel Adv	**0.89**									
Compat	0.44	**0.95**								
Compl	0.16	0.20	**0.87**							
Trial	0.35	0.10	0.25	**0.91**						
Observ	0.15	0.11	0.02	0.29	**0.91**					
ATB	0.42	0.35	0.06	0.19	0.37	**0.88**				
SN	0.25	0.15	0.11	0.29	0.32	0.18	**0.93**			
EO	0.42	0.17	0.33	0.70	0.10	0.16	0.27	**0.86**		
PBC	0.60	0.23	0.35	0.75	0.22	0.25	0.31	0.80	**0.89**	
INT	0.60	0.21	0.30	0.70	0.27	0.18	0.40	0.83	0.80	**0.92**

Abbreviation: Rel Adv, Relative advantage. Compat, Compatibility. Compl, Complexity. Trial, Trialability. Observ, Observability. ATB, Attitude to behaviour, SN, Subjective norms. EO, Entrepreneurial Orientation. PBC, Perceived behavioural control. INT, Intention to adopt.
Source: Survey data.

The results of discriminant validity are shown in Table 6. The correlation matrix of latent exogenous constructs is all below the permissible level (0.85), and the square root of AVE (diagonal values in bold) for each construct is larger than the correlation between corresponding constructs, implying that the measurement model achieves discriminant validity (Ahmad *et al*., 2016). The model's convergent validity was attained, as expected because the construct parameter estimations are significant and coherent with the theory. Similarly, according to the goodness of fit (GOF) indices (Table 4.7), the model fits well and achieves the level of acceptability, indicating construct validity as specified by Hair *et al*. (2010). Finally, the measurement model's validity and reliability were satisfactory. Therefore, the structural model was assembled for further analysis.

4.5 Prediction of farmers' attitudes and intention to adopt or continue adopting CSA

Table 4. 7: Model fit indices for the unconstrained and constrained models

Statistic	Threshold	Unconstrained model	Constrained model
CMIN/DF	1-3	2.023	2.026
TLI	≥0.90	0.929	0.929
CFI	≥0.90	0.938	0.937
RMSEA	<0.8	0.067	0.067
SRMR	<0.09	0.050	0.050
χ^2**,df,*p***	-	χ^2 =1127.116, df= 557, *p*=0.000	χ^2 = 1126.664, df= 556, *p*=0.000

Source: Survey data. Thresholds adapted from Hair *et al*. (2010)

Table 4. 8: Chi-square test for comparison between the constrained and unconstrained model

Model	Chi-square	df	*P*-value
Unconstrained	1127.116	557	
Constrained	1126.664	556	
Difference	0.45205	1	0.5014

Source: Survey data.

The Chi-square test used to compare the two models yielded an insignificant result (p-value = 0.5014), showing that the two models are not significantly different (p-value = 0.5014) (Table 4.8). This result also means that the constrained structural model can better describe the constructs' influences on farmers' intentions to adopt CSA technologies.

Table 4. 9: Structural Equation Modelling parameter estimates(coefficient) for farmers' attitude and intention to adopt or continue adopting CSA

Endogenous variable	Predictors	Coefficient	SE	Z	*p* value
Intention to adopt	Attitude	0.074	0.025	2.925	0.003**
	Subjective norm	-0.084	0.038	-2.215	0.027**
	Perceived behavioural control	1.548	0.264	5.876	0.000***
	Entrepreneurial orientation	0.573	0.268	2.140	0.032**
Attitude	Relative Advantage	0.275	0.071	3.843	0.000***
	Compatibility	0.157	0.068	2.301	0.021**
	Complexity	-0.031	0.073	-0.420	0.675
	Trialability	0.010	0.083	0.124	0.902
	Observability	0.269	0.066	4.078	0.000***

** Statistically significant at 5%. *** Statistically significant at 1%. Source: Survey data.

The structural equation modelling estimation for the intention to adopt or continue adopting CSA technologies and the attitude toward CSA technologies are summarized in Table 4.9. In the case of the former, all analyses reported statistically significant parameter estimates. The theory of planned behaviour constructs (Attitude, Perceived behavioural control, and subjective norms) and the entrepreneurial orientation both had significant coefficients, as expected. The more positive a coffee farmer's attitude toward CSA technologies is, the more likely they will adopt them. The greater the sense of control coffee growers have with CSA technology, the more likely they will adopt them (most significant).

In contrast, a negative relationship exists between subjective norms and the intention to adopt or continue adopting CSA technologies. The more CSA technologies clash with society's subjective norms, the less likely farmers will adopt them. However, the more entrepreneurially oriented the coffee farmers are, the more they intend to adopt CSA technologies. Regarding attitude predictors, Relative advantage, compatibility and observability are significant and positively influence the farmers' attitude towards CSA technologies. As expected, complexity negatively influences farmers' attitudes towards CSA technologies; however, it is not significant just as trialability.

4.6 Adoption of Climate-Smart Agriculture technologies

We classified CSA practices based on the principal resource required, using Amadu *et al*. (2020) CSA typology (Appendix 2). The first group includes primarily labour-intensive technologies. Farmers must conduct basic operations such as mechanical weeding, pruning, de-suckering, cultural pest and disease control, cover crops, gap filling, stumping, and intercropping as part of CSA practices in this category. Practices that primarily demand land use rights fall into the second group. Physical structures for soil and water conservation, as well as the planting and management of shade trees, require farmers to own the land entirely. Practices that primarily require financial capital fall into the last group. Farmers must pay for these techniques, such as improved varieties, chemicals, mulching, manure application, irrigation and inorganic fertilizers. We overcome the difficulty in categorizing objects by focusing on the essential resource, i.e. the resource required in high quantities, without which the practice cannot be implemented.

This method of categorizing CSA activities has lately been employed in Kenya (Kangogo *et al*., 2021). This procedure has resulted in a categorisation based on three types of indispensable resources; Land, labour and capital. CSA technology, which farmers wish to continue adopting (75.5%), is linked to labour as a requirement, and the intention to adopt is no different at 80.2% for labour-intensive technologies. Mechanical weeding technology, in particular, receives the highest scores in the labour category, with 100% continued adoption and 98.7% intention to adopt. However, cover crops receive the lowest scores in the labour category, with 41.70 % continued adoption and 52.20 % intention to adopt. It is unsurprising that CSA technologies that require capital have the slightest intention of being adopted (49.2%) and sustained (26.4). In comparison, irrigation has the lowest continuing adoption and intention to adopt rates at 7.40 % and 19.10 %, respectively. Almost every practice has a higher percentage of coffee growers who intend to adopt it than those currently using it.

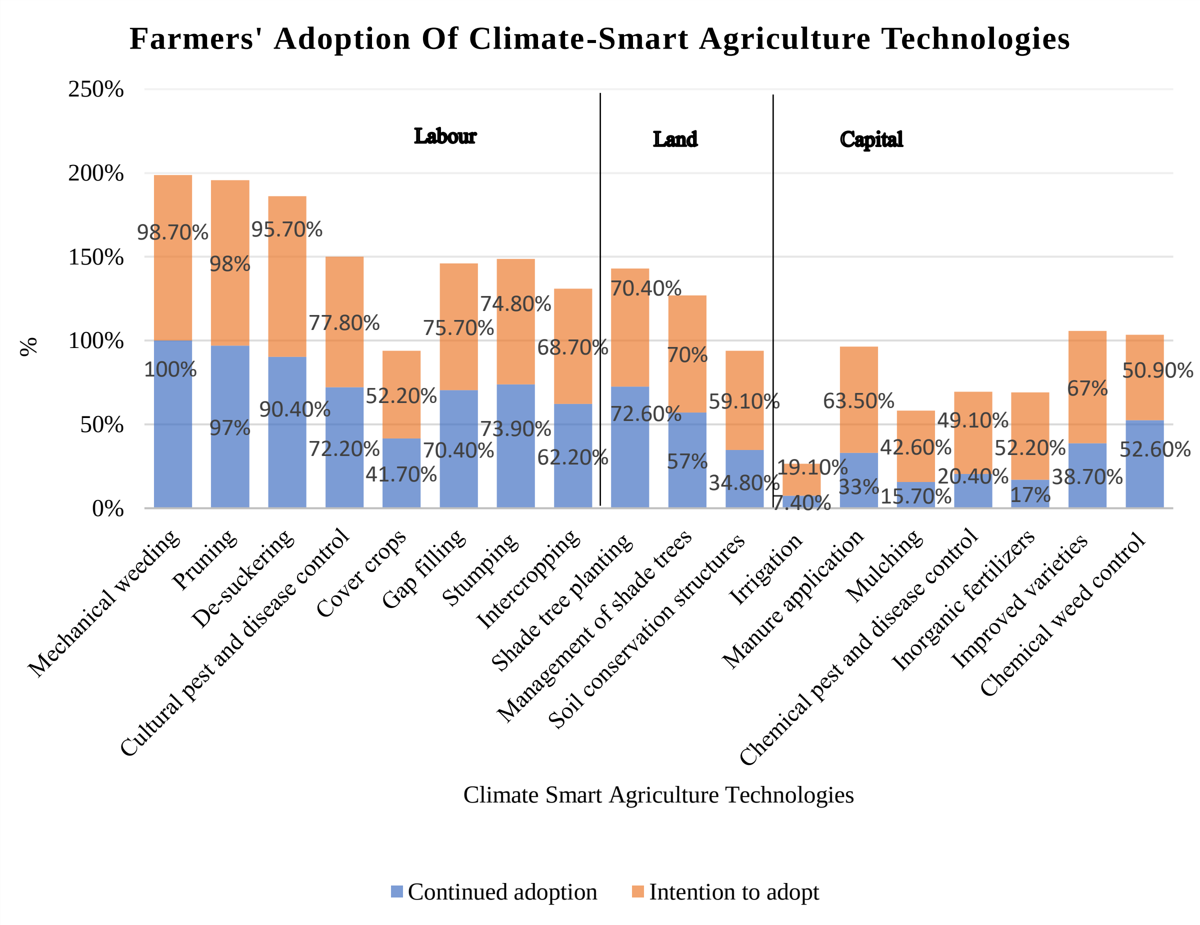


Figure 4- 3:Evaluation of farmers' adoption of individual CSA technologies. Percentages were based on the total number of respondents, N=230

4.7 Adoption across entrepreneurial mindset clusters

Except for capital-intensive technologies, most farmers' intentions to adopt and continued adoption is over 50% for all technologies, irrespective of the entrepreneurial mindset (figure 4.4). It is unsurprising that most of the variation in adoption intention occurs with technologies in the category of capital intensive technologies, with enterprising farmers having the highest intention to adopt capital intensive technologies (87%) and 51% on the side of continued adoption. It is no wonder non-enterprising farmers have the lowest intention to adopt capital-intensive technologies at 55% and 35% for continued adoption. The eight technologies that fall under the labour category obtain the highest intention to adopt and continued adoption rates (66% - 76%) and (68% - 70%), respectively, across all entrepreneurial mindset clusters. It is not surprising that non-enterprising and indecisive coffee farmers have the highest intention to adopt CSA technologies that are more labour intensive at 76% and 75%, respectively and 68% and 68% for continued adoption, respectively. Also, the CSA technologies that require land rights are overly intended to be adopted by non-enterprising farmers (73%) and 59% for continued adoption. Conservative farmers have the slightest intention to adopt (61%) in this category and 57% for continued adoption.

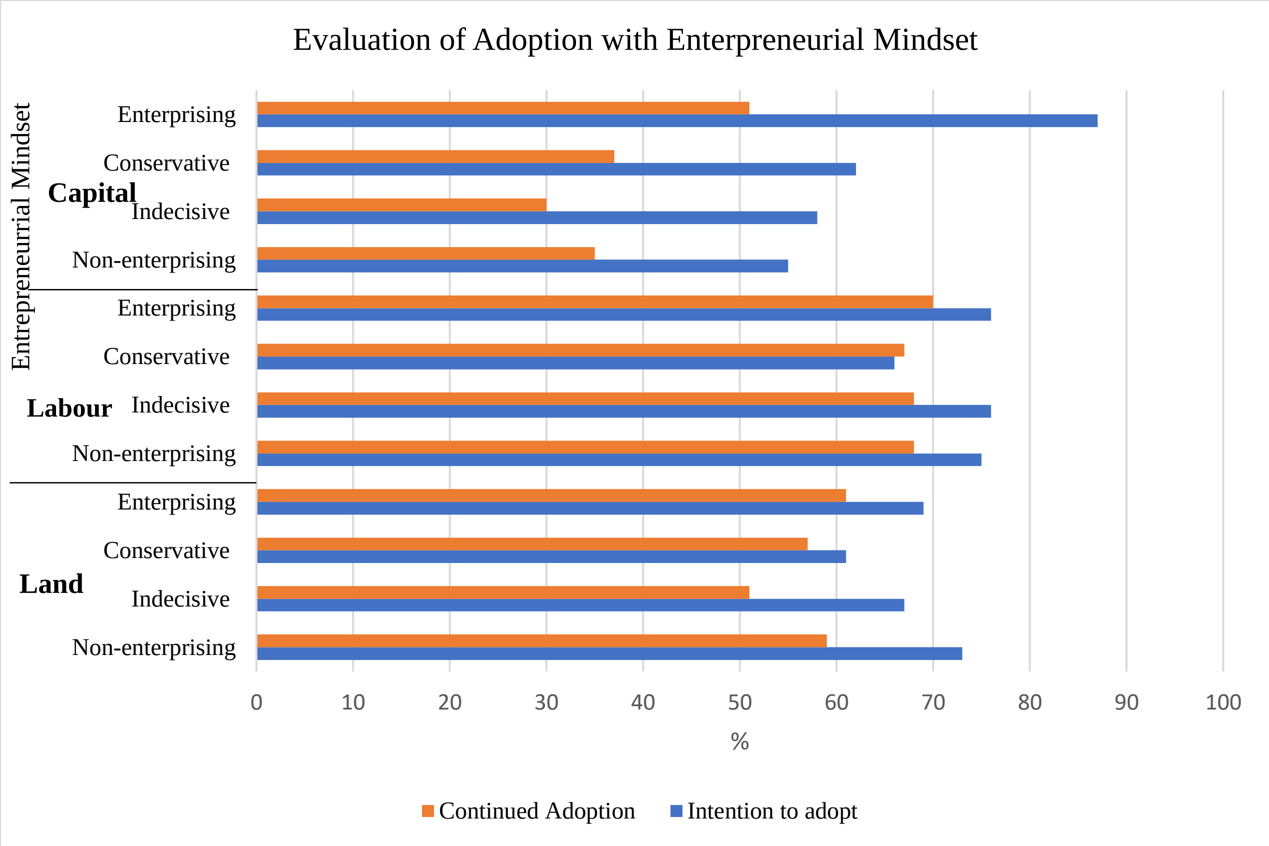


Figure 4- 4:Famers' intention and continued adoption of Climate-Smart Agriculture technologies depending on the essential requirements per entrepreneurial mindset cluster. Percentages were based on the total number of respondents, N=230

CHAPTER FIVE: DISCUSSION

5.0 Introduction

This chapter discusses the research findings in detail, including the interpretation, implications, limitations, and the study's contribution to knowledge and conclusions. The chapter ends with policy and practice recommendations and suggestions for further research.

5.1 Socio-demographics and Farm Characteristics of Respondents

Study findings revealed that most participants were men (67.8%), while females made up 32.2% which deviates from the country ratio of almost 1:1 (UBOS, 2021). This result indicates that men participate more actively in growing coffee than women, as they own more land than women. These findings are consistent with those of ICO (2018), which found a gender bias in coffee farming, with male farmers dominating the industry. Because women are crucial in the actual farm operations, this could deny female farmers active engagement in coffee farming activities, limiting coffee production. Coffee growing is generally carried out by older adults over the age of 50, as seen by the average age of coffee farmers (52.6 ±14.8 years). This finding is in line with Aguirre *et al*. (2022), according to which the average age for coffee producers was 55 years. Ngeywo *et al*. (2015) also found that 71% of those involved in coffee production were above 50 years. As a result, youth involvement in coffee production is limited, posing a risk to the growth and continuity of the coffee industry. Younger farmers are more likely to adopt innovative farming techniques, whereas elderly farmers reject energy-intensive technologies (Kagezi *et al*., 2021).

The average household size of coffee farmers (8±5 members) is higher than the country's average (5 members) (UBOS, 2021). This finding can be attributed to the relatively high fertility rates among rural Ugandan women and a low prevalence of contraceptive use due to cultural preference for large families, which associate children with prestige and source of labour (Sserwanja *et al*., 2021). Our study found that the average years of formal education is 6.40±4.8 years. This finding indicates that coffee farmers attained basic primary education, which supports the prevailing evidence that the highest education level attained by most adults in Uganda is 1-7 years (UBOS, 2021). This evidence is attributed to the government's free education through Universal Primary Education.

5.2 Information about Climate-Smart Agriculture and Business Training

Regarding climate-smart agriculture information, our research indicated that most respondents do not have access to CSA information, necessitating more communication efforts when deploying new technologies. CSA was adopted in Uganda and incorporated in the Agriculture Sector Strategic Plan of 2015/16–2019/20 (USAID & CIAT, 2017). Surprisingly, coffee farmers have no access to CSA information. However, the government has extension workers recruited in every district, and the extension channels could be utilized to create awareness about CSA. Our findings support the assertion that information access in Uganda is still limited and that further efforts are needed to raise awareness (UBOS, 2021). Although most respondents stated that they had not attended business training, this does not imply that coffee farmers are unaware of how to grow and sell coffee. Instead, they do so voluntarily as a traditional cash crop with little or no attention paid better to managing their farms' business and agricultural aspects. Our findings back up the existing data that farmers have limited access to training and hence have more significant training needs (Adeogun *et al.*, 2019; Edidah *et al.*, 2020). On the other hand, the low attendance of business training can be attributed to the respondents' low educational level, which limits them.

5.3 Segmentation of Coffee Farmers According to their Entrepreneurial Orientation

Cluster analysis revealed variation among coffee farmers and four levels of entrepreneurial mindset: non-entrepreneurial, indecisive, conservative, and enterprising. The bulk of coffee farmers (40.9 %) are classified as "indecisive" and have a low entrepreneurial orientation score (risk-taking, innovativeness and proactiveness). Farmers' age, years of agricultural experience, total farmland, and monthly income were significant drivers of cluster membership, consistent with earlier research (Jassogne *et al*., 2017). Coffee farmers with the highest monthly income and largest acreage of land cultivated are significantly more represented in the “enterprising” cluster. Our findings support prior research that farmers with more farmland are more enterprising than their peers (Suvanto *et al*., 2020). This result is attributed to the high number of years of coffee farming and diversity in production. Regarding attention paid to business model canvas (BMC) elements, customer segments, value propositions, revenue streams, and cost structures are vital determinants of cluster differentiation. Coffee farmers who pay most attention to customer segments and value propositions are more represented in the “enterprising” cluster, and the reverse is true for the “non-enterprising” cluster. Revenue streams and cost structures play different roles in each cluster. Coffee farmers who pay the most attention to revenue streams are more represented in the non-enterprising cluster. However, they significantly pay the slightest attention to cost structures due to their low levels of record-keeping as for most farmers in developing countries (Navarro *et al*., 2020). Moreover, lack of documentation and record-keeping is a crucial determinant of the lack of sustainability of the farm businesses.

5.4 Coffee Farmers’ Perceived Compliance to Business Model Canvas (BMC) elements

Our findings back up previous research on the most significant business model elements (Franceschelli *et al*., 2018; Metallo *et al*., 2018; Müller, 2019). Specifically, the value propositions, customer segments, communication channels, and customer relations are the elements that farmers believe have a wealth of information and pay close attention to. Nevertheless, coffee farmers believed they pay the slightest attention to the revenue streams, followed by key partners and cost structures. This finding can also be shown in the cluster analysis, where the perceived attention paid to customer segments and value propositions is high and significantly different across farmer clusters. Similarly, the perceived attention paid to revenue streams and cost structures is low and significantly different across entrepreneurial mindset groups. Even though studies on the internet (Metallo *et al*., 2018) and small and medium enterprises (Müller, 2019) show that key resources and key partners are the most significant BMC blocks, our findings differ since coffee farmers perceive they pay modest attention to the two elements. This result supports the assertion that business models are a unique feature of value configuration in firms and differ significantly among firms and sectors (Joyce & Paquin, 2016; Teece, 2018). Therefore, future research might look into the business model deployed by coffee farmers when adopting CSA technologies.

5.5 Predicting Attitude of Coffee Farmers Towards Adoption of Climate-Smart Agriculture technologies

Structural Equation Modelling (SEM) revealed the influence of DOI's perceived technology characteristics on the attitude of coffee farmers towards CSA adoption. Concerning the five perceived technology characteristics, relative advantage, compatibility and observability are the only statistically significant predictors of attitude. Moreover, relative advantage and observability are the most significant predictors of attitude towards CSA adoption. This outcome corresponds with the findings of Chou *et al*. (2012). They found that perceived technology characteristics positively and directly influence attitudes towards green practices and behavioural intention to adopt green practices. In contrast, trialability and complexity do not significantly influence coffee farmers' attitudes towards CSA adoption. Relative advantage affects attitude positively, suggesting that the more superior CSA technologies are to existing technologies, the more favourable the attitude of coffee farmers toward adopting them. Similarly, the observable and compatible CSA technologies are the more positive the attitude of coffee farmers towards adopting them. Our findings corroborate the results of similar research that combined DOI and TPB (Jung Moon, 2020) and studies that used DOI perceived innovation characteristics as the single predictor of attitude (Putzer & Park, 2012). Therefore, communication efforts are needed to create awareness about particular perceived technology characteristics, which are critical in forming a positive attitude toward CSA adoption. When respondents believe CSA adoption is similar to existing technologies, they have a favourable attitude towards their adoption. Compared to other agricultural technologies, CSA is regarded to have a higher relative advantage in terms of economic, technical, and environmental elements. The more farmers perceive CSA as observable by others, the more positive attitudes regarding CSA adoption develop.

5.6 Predicting the Intention to Adopt or continue adopting Climate-Smart Agriculture technologies by Coffee Farmers

Our findings on coffee farmers affirm the growing body of literature on adopting climate-smart agriculture technologies in the coffee industry (Buyinza *et al*., 2020). According to the model estimates, the TPB constructs of "attitude" and "perceived behavioural control" had a significant positive influence on farmers' intentions to adopt and continue adopting CSA. However, "subjective norms" had a significant negative influence. The cognitive construct of "entrepreneurial orientation" was added as a relatively novel yet relevant component that positively and significantly influenced coffee farmers' intention to adopt and continue adopting CSA.

Concerning coffee farmers' attitude towards CSA adoption, our results are similar to several studies that noted a significant and positive influence of attitude on behavioural intention (Cristea & Gheorghiu, 2016; Dalila *et al*., 2020; Hasan & Suciarto, 2020; Mead & Irish, 2021;

Shan *et al*., 2020). This finding means that coffee farmers with a positive attitude towards CSA are more likely to adopt CSA technologies on their farms than their counterparts. Although several studies have reported attitude as the strongest and most consistent predictor of behavioural intention (Chen, 2017; Prati *et al*., 2012), our results differ since perceived behavioural control is the most significant predictor of farmers' intention to adopt CSA. Nevertheless, our results are consistent with Buyinza *et al*. (2020), who reported that perceived control is the most significant predictor of behavioural intention to adopt shade trees in coffee farms. Coffee farmers who perceive a high level of control over CSA implementation are more likely to adopt the technology than those who believe they have little or no control. Farmers typically argue, according to Buyinza *et al*. (2020), that deploying CSA technologies on their farms is out of their control without significant support from different actors like organizations and the government to remove impendiments such as lack of improved seedlings, inadequate tools, and access to water. As such, a multistakeholder approach is key in scaling up the adoption of CSA technologies by coffee farmers.

In terms of subjective norms, our findings contrast several studies that found a significant and positive influence on intention (Dalila *et al*., 2020; Ibrahim & Arshad, 2017; Lee, 2011; Mead & Irish, 2021). Our results are, however, consistent with Cera & Furxhiu (2017); Hasan & Suciarto (2020); Smith (2015), who found a significant negative relationship between Subjective norms and behavioural intention. This outcome implies that CSA technologies will be less likely to be adopted on coffee farms if they undercut or clash with pre-existing social-cultural attachments. As such, CSA technologies must be context-specific to match the social norms but also must be introduced using participatory approaches and communicated through social channels and structures such as community leadership for the technologies to be widely adopted.

Regarding entrepreneurial orientation (EO), enterprising coffee farmers are more likely to adopt or continue adopting CSA technologies on their farms than their counterparts. Unlike earlier studies that looked at EO from a multidimensional perspective (Kangogo *et al*., 2021), our study looked at EO from a single dimension to assess its overall impact. Our findings support earlier research that suggests that entrepreneurship influences farmers' productivity and technological decisions (Passarelli *et al*., 2021; Suvanto *et al*., 2020). EO is critical for identifying opportunities and overcoming the financial limitations that hinder farmers from adopting CSA technologies. Furthermore, Pindado *et al.* (2018) claim that innovative farmers have a competitive edge and are more sustainable in modern agriculture. As a result, coffee farmers should undertake business and entrepreneurship skills training to prepare them to embrace new technology for sustainable production in the face of climate change.

5.7 Adoption of Climate-Smart Agriculture Technologies

Regarding CSA technology adoption, our findings confirm that coffee farmers are slow to adopt the technology (Bunn *et al*., 2019; USAID & CIAT, 2017). Unlike earlier studies that look at CSA technologies individually (Nakabugo *et al*., 2019), we classified CSA technologies based on the basic requirements for adoption to assess the impact of production factors on coffee farmers' adoption intentions. Coffee farmers are more likely to adopt CSA technologies that primarily demand labour (80%) than those that require land (66.5%) and capital (49.2). Although coffee farmers tend to adopt labour-intensive CSA technologies (Kangogo *et al*., 2021), 47.8% of coffee farmers do not intend to adopt cover crops. This finding may be due to the cost of buying and managing cover crops, even though it is not a critical requirement for the technology (Dunn *et al*., 2016).

Similarly, just 19.10% of coffee farmers intend to adopt irrigation, a low percentage compared to the average of farmers intending to adopt capital-intensive technologies (49.2%). This outcome is due to rural areas' limited availability of water, which raises the capital requirements for a farmer to build their groundwater supply (Nakawuka *et al*., 2018). The whole situation implies that capital constraint as a crucial production factor is prevalent among coffee farmers. Hence increasing credit availability to farmers is critical to supporting their attempts to adopt CSA technologies. This credit could be in the form of cash or inputs.

Furthermore, farmers were segmented based on their entrepreneurial mindset to target a specific farmer cluster with an appropriate technology set. First, irrespective of the entrepreneurial mindset cluster, all coffee farmers intend to adopt CSA labour-intensive technologies in significant numbers. Notably, non-enterprising and indecisive coffee farmers' adoption intention rates are 76% and 75%, respectively. This result is because most rural farmers rely on family labour (Kangogo *et al.*, 2021), as seen by the high average number of household members (Table 4.1). Labour-intensive CSA technologies are well-known routine practices associated with low costs and little risk. This minimal hindrance makes non-enterprising and indecisive farmers adopt with comfort.

Second, concerning land-related CSA technologies, all entrepreneurial mindset clusters have adoption intention rates higher than 61%. This outcome can be attributed to the fact that coffee is a perennial crop and most coffee farmers plant it on their land (Jassogne *et al*., 2017). Hence, land rights are not a significant limitation for adopting such technologies. Moreover, such technologies are less costly as they involve establishing permanent structures (trees and trenches), which are done once and only maintained occasionally.

Finally, enterprising coffee farmers are most likely to adopt capital-related technologies (87%). Their interest could be a result of new technology and innovation that has the potential to drastically alter the production system, as is typical of capital-related technologies (Suvanto *et al*., 2020). Earlier research has shown that enterprising farmers are more likely than their peers to adopt capital-intensive technologies (Amadu *et al.*, 2020; Kangogo *et al*., 2021). In so doing, enterprising coffee farmers are fast at identifying opportunities, and they overcome financial barriers. Unfortunately, in sub-Saharan Africa, farmers' access to entrepreneurial training is limited since they are not integrated with the agronomic training farmers receive (Kangogo *et al*., 2021). Therefore, to enhance the adoption of CSA technologies, extension processes should be designed to deliver entrepreneurship training alongside agronomic training but also consider variances in coffee farmers' entrepreneurial mindsets (Bunn *et al*., 2019). This recommendation could be achieved through farmer field schools where different farmer groups undergo experiential and experimental learning (Chandra *et al*., 2017).

5.8 Significance of the Study

The current adoption study offers vital information that many stakeholders who may be involved in the adoption and sustainable scaling of climate-smart agriculture technology need. First, it shows that, in addition to the Diffusion of Innovation and the Theory of Planned Behavior, the Entrepreneurial Orientation theory is a viable one that researchers working on the creation and sustainable scaling of CSA technologies can use to comprehend farmers' motivations for accepting these technologies. Second, the study provides information that policy makers can use to guide their choices for actions to promote the acceptance and sustainable scaling of CSA technologies. Third, the study informs farmers about the potential of climate-smart agriculture to increase food security and climate resilience while also reducing greenhouse gas emissions.

5.9 Limitations and Future Research

First, the validity of the Likert scale measurement of model constructs might have been compromised because the mean values were generally high. Because people like to think positive of themselves, social desirability has an impact on how people use the Likert scale (McLeod, 2008). To minimise this effect the study provided anonymity on questionnaires to minimize social pressure and, in turn, reduce social desirability bias. According to Paulhus (1984), participants reported more attractive traits about themselves when they were asked to fill out their names and contact information rather than when the questionnaire was anonymous.

Second, the study's geographical scope is limited because it was conducted in only one coffee-growing region in Uganda (greater Luwero). The adoption of farming technologies is influenced by institutional factors, land tenure, resource availability, economic, social, ecological and climate conditions (Engel & Muller, 2016; FAO, 2016; Khatri-Chhetri *et al*., 2017; Mwongera *et al.*, 2017). These factors vary across regions, so it is important to interpret our results with caution, and they need to be validated in future research to strengthen generalizability. Third, our study analyzed entrepreneurial orientation from a single dimension perspective whereas other studies acknowledge the multidimensional perspective involving different components of risk-taking, innovativeness and proactiveness (Kangogo *et al.*, 2021; Suvanto *et al*., 2020). Future research needs to consider the multidimensional perspective of EO and how each component influences farmers' intentions to adopt or continue adopting CSA technologies by coffee farmers.

Fourth, the study was a cross-sectional survey based on data collected at a specific time. However, the evolution of farmers' entrepreneurial competencies informs the changes in their adoption behaviour (Pindado & Sánchez, 2017b). Panel data should be collected in future research to understand the formation process of farmers' entrepreneurial behaviour and the efficacy of interventions to enhance farmers' entrepreneurial mindset. Lastly, our analysis is based on farmers' intention to adopt or continue adopting CSA technologies but not on actual adoption behaviour. Future research could explore the adoption of CSA technologies by coffee farmers and examine the relationship between farmers' intentions to adopt CSA technologies and actual adoption (Kupfer *et al*., 2016). Adoption could also be studied at different stages (potential adopters Vs experienced users) over time (Kim, 2008). Furthermore, it may be relevant for future research to conduct an impact analysis on adopting CSA technologies. Findings on how CSA technologies impact coffee yield, household livelihoods and GHG emissions are important for influencing farmers' attitudes towards promoting adoption (Asegid, 2020; Kearney *et al.*, 2017; Sardar *et al.*, 2020; Sun *et al.*, 2018).

6 Conclusion

Our study examined coffee farmers' entrepreneurial mindsets and the factors influencing their intentions to adopt or continue adopting climate-smart agriculture technologies based on a survey of 230 coffee farmers in greater Luwero, Uganda. Coffee farmers' attitudes towards the adoption of CSA technologies were found to be predicted by DOI's perceived technology characteristics. The most important predictors were relative advantage and observability, with complexity as a potential predictor of negative attitudes. TPB constructs (attitudes, perceived behavioural control, and subjective norms) also merged as significant predictors of coffee farmers' intentions to adopt and continue adopting CSA technologies. Meanwhile, subjective norms merged as a potential barrier to CSA technology adoption, with perceived behavioural control merged as the most significant predictor. Finally, coffee farmers' intentions to adopt or continue using CSA technologies were found to be significantly influenced by their entrepreneurial orientation. These findings added to the existing literature a valuable factor (EO) for predicting farmers' adoption intentions.

Our cluster analysis revealed a small percentage of enterprising coffee farmers (20.4%) and 79.6 % of coffee farmers in other clusters (40.9% indecisive, 25.2% conservative and 13.5% non-enterprising), confirming the low entrepreneurial mindset of coffee farmers. Furthermore, non-enterprising farmers have the smallest amount of farmland and the lowest monthly household income. Interventions could focus on the most overrepresented coffee farmers in this cluster, namely the youngest coffee farmers (42 ±14 years) with little experience in agriculture (20 ±15years) and coffee farming (18 ±16years). Although resources limit coffee farmers' ability to adopt CSA technologies (Bunn *et al.*, 2019; USAID & CIAT, 2017), our extensive technology study revealed new insights on the intention to adopt or continue adopting CSA technologies based on the essential requirement and across various entrepreneurial mindsets clusters. On average, nearly all farmers (80%) intend to adopt labour-intensive CSA technologies and just 65% for land-related technologies. In contrast, only 49.2% intend to adopt capital-intensive technologies, of which 87% were enterprising coffee farmers. The low scores on entrepreneurial orientation, information and attention paid to business model canvas elements should be tackled to increase coffee farmers' entrepreneurial mindsets, which will increase their intentions to adopt CSA technologies.

Thus our study suggests that efforts to scale up the adoption of CSA technologies in the study areas should focus on; 1) Raising awareness about the perceived characteristics of CSA technologies to influence positive attitudes of farmers towards adopting and sustaining CSA; 2) Employing a multistakeholder approach to remove impediments that hinder adoption by coffee farmers; 3) Providing business and entrepreneurial skills training for coffee farmers and extension programs should be designed to take into account variances in the farmers' entrepreneurial mindsets.

This research makes a two-fold contribution to the debate over technology adoption. First, the paper expands the literature on adoption factors beyond generic factors like institutional, behavioural, land tenure, resource availability, economic, social, ecological and climate conditions (Engel & Muller, 2016; FAO, 2016; Khatri-Chhetri *et al*., 2017; Mwongera *et al.*, 2017). It includes novel insights from cognitive traits like entrepreneurial orientation, a new concept in agricultural research. Second, the analysis links the four entrepreneurial mindset clusters to the CSA categories based on the critical resource required for adoption, using the typology defined by Amadu *et al*. (2020). Examining how farmer entrepreneurial mindset and resource constraints influence CSA adoption offers a solid platform for more targeted sustainable scaling policies and intervention strategies to be designed and implemented.

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APPENDIX I: QUESTIONNAIRE

Informed consent

Dear Respondent,

I am Senyange Brian, a Ghent University, Belgium student undertaking an International Master's of Science in Rural Development. In partial fulfilment of the requirements for an International Master's of Science in Rural Development Degree award, I am undertaking research titled Farmer's Entrepreneurial Mindset and the Intention to Adopt Climate-Smart Agriculture: The Case of Coffee Farmers in Greater Luwero, Uganda. As part of fulfilling the requirements for conducting research, I would like to ask you whether you accept to participate in this research freely. The research respondents are smallholder coffee farmers, and you have been randomly selected to participate in the research. A digitized structured questionnaire will be used in the study. You will be asked the different questions in the questionnaire by a trained research assistant privately on a one-on-one basis. The research assistant will input your responses to the different questions, and your responses will be recorded anonymously without indicating your names. There are no risks involved in partaking in the study.

Participation in this survey is voluntary. You are free to refuse to participate before the study begins, discontinue at any time, or skip any questions that may make you uncomfortable. You are also allowed to ask questions about the study before agreeing to be involved and during the study. Your names will not be noted in the responses. Instead, the pre-determined unique numeric ID will be used to ensure confidentiality. In the end, the enumerator will upload the responses to the ODK server immediately. They will not be shared with any other member.

Respondents' consent

Yes, I consent to take part in the research

No, I do not consent to take part in the research

SECTION A:

SOCIO-DEMOGRAPHICS AND FARM CHARACTERISTICS

1. Gender. Male / Female
2. How old are you? ……………………………
3. How many years of education do you have? …………………………..
4. What is your Marital status? …………………………..
a) Single b) Married c) Divorced/Separated d) window(er)
5. How many years of education do you have? …………………………
6. How many people live in your household? …………………………..
7. Are you the household head? Yes/No
8. How many years have you spent practising agriculture? …………………………
9. How many years have you spent in coffee farming? ……………………………..
10. What is the total size of land you own (acres)?...

11. Of the total land, how much do you hire in (acres)?..
12. What type of livestock do you own?
a)Sheep b)Cow c)Goats d)Chicken e)Rabbits f)Pigs g)Turkey h)Ducks
13. Have you ever attended any business skills training? Yes/No
14. What is your monthly household income (UGX)?...
15. Have you received information on CSA technologies? Yes/No

Information regarding Climate-Smart Agriculture

CSA refers to agricultural techniques with the potential to increase production to support increased revenues, food security, and development; Build climate resilience; Reduce Green House Gas emissions; Improving carbon sequestration. CSA has several technologies, which include; integrated soil fertility management (Using both inorganic and organic fertilizers); Shade tree planting and management; Intercropping coffee-banana; Cover cropping; Mulching

SECTION B: DOI COMPONENTS

13. For the following statements, please indicate your opinion to what extent you agree or disagree with them by ticking (√) where applicable. Note 1= strongly disagree, 7 = strongly agree

Variable	Opinion statement	Indicate your response by ticking one of the options below						
		1	2	3	4	5	6	7
-Relative advantage	Practising CSA can enable me to accomplish farm work more easily							
	Practising CSA can improve the quality of my coffee produce							
	Practising CSA can make it easier to grow coffee							
	Practising CSA can enhance my effectiveness in coffee production							
	Practising CSA can give me greater control over my coffee farm							
Compatibility	Practising CSA can match the traditional ways of operating the coffee farm							
	Practising CSA can fit well the way I like to do farming							
	Practising CSA can fit into my farming style							
Complexity	I believe that it is easy to practice CSA for a particular purpose							
	Overall I believe that Practising CSA is easy							
	Guideline for Practising CSA can be easy for me							
Trialability	Before deciding on whether to practice CSA, I need to try them out properly							
	It can be easy for me to practice CSA if I am permitted to use it on a trial basis long enough to see what it can do							
	Practising CSA on a trial basis may be easy for me							
Observability	CSA has been well demonstrated in my village							
	I believe that people can tell that your farm has changed when you practice CSA							
	I believe that practising CSA can improve the way my farm looks like							

SECTION C: TPB COMPONENTS

14. For the following statements, please indicate your opinion to what extent you agree or disagree with them by ticking (√) where applicable. Note 1= strongly disagree, 7 = strongly agree

Variable	Opinion statement	Indicate your response by ticking one of the options below						
		1	2	3	4	5	6	7
Attitude	Practising CSA in part of my coffee farm next year is advantageous							
	Practising CSA in part of my coffee farm next year is necessary							
	Practising CSA in part of my coffee farm next year is important							
Subjective norm	Most people who matter to me would support me to practice CSA with my coffee farm next year							
	Most people whose opinion I value would approve that I practice CSA within my coffee farm next year							
	Most farmers like me will practice CSA in at least part of their coffee farm next year							
Perceived Behavioral control	I have sufficient resources to practice CSA in part of my coffee farm next year							
	I am confident that I can practice CSA on the part of my coffee farm next year							
	Deciding on whether to practice CSA on my coffee farm within next year is entirely up to me							

SECTION D: INTENTION TO ADOPT CSA

15. For the following statements, please indicate your position concerning the statement on the scale of 1-7;1=unlikely, 7= Most likely

Variable	Opinion statement	Indicate your response by ticking one of the options below						
		1	2	3	4	5	6	7
Intention to Adopt CSA	How likely is it that you will practice CSA in part of your coffee farm within the next year							
	I intend to practice CSA in part of my coffee farm within the next year							
	I may practice CSA in part of my coffee farm within the next year							

16. Which CSA technologies are you using on your coffee farm right now, and those you are willing to practice next year

CSA practices	**Period**	
	Right now	**Next year**
Mechanical weeding		
Pruning		
De-suckering		
Manure application		
Mulching		
Cultural pest and disease control		
Chemical pest and disease control		
Inorganic fertilizers		
Shade tree planting		
Management of shade trees		
Cover crops		
Irrigation		
Improved varieties		
Gap filling		

Stumping		
Intercropping		
Soil conservation structures		
Chemical weed control		

SECTION E: EO COMPONENTS

17. For the following statements, please indicate your position by circling on the 1-7 continuous scale linking the opposite statements

Opinion statement	Scale	Opinion statement
Risk-Taking On my farm, I prefer to follow;		
Established practices and routines	1 2 3 4 5 6 7	Possibilities to change practices and routines
Low-risk farming practices	1 2 3 4 5 6 7	High-risk farming practices
For decision-making involving uncertainty,		
I typically adopt a cautious "wait and see" posture to minimize making costly decisions	1 2 3 4 5 6 7	I typically adopt an aggressive posture to maximize the possibility of exploiting potential opportunities
Innovativeness In general, I favour a strong emphasis on;		
Well-known agricultural practices and production systems	1 2 3 4 5 6 7	R&D, new technologies and innovation
How many new crops or services has your farm produced in the past five years		
No new crops or services	1 2 3 4 5 6 7	Many new crops or services
Changes in the production system and practices have been minor	1 2 3 4 5 6 7	Changes in the production system and practices have been dramatic
Proactiveness Our relationship with similar farming businesses is characterized by;		
Is seldom the first farm to adopt new crops and practices	1 2 3 4 5 6 7	We are very often among the first farms to adopt new crops and practices
Typically respond to actions/practices initiated by other farms	1 2 3 4 5 6 7	Typically initiates actions or practices that other farmers respond to

SECTION F: BMC

18. For the following statements, please indicate your opinion to what extent you agree or disagree with them by ticking (√) where applicable

Opinion statements	Scale				
	1	2	3	4	5
I have enough information regarding the following elements of my farm					
Revenue streams					
Cost structures					
Value propositions					
Key resources					
Channels					
Customer relations					
Customer segments					
Key partners					
Key activities					
Please indicate to what extent you pay attention to the following elements in your farm. 1=No attention, 5=Much attention					
Revenue streams					
Cost structures					
Value propositions					
Key resources					
Channels					
Customer relations					
Customer segments					

Key partners					
Key activities					

Thank you for participating in this study

APPENDIX II: CATEGORIES OF CSA PRACTICES

CSA practices	**CSA category**		
	Labor	**Land rights**	**Capital**
Mechanical weeding	✓		
Pruning	✓		
De-suckering	✓		
Manure application			✓
Mulching			✓
Cultural pest and disease control	✓		
Chemical pest and disease control			✓
Inorganic fertilizers			✓
Shade tree planting		✓	
Management of shade trees		✓	
Cover crops	✓		
Irrigation			✓
Improved varieties			✓
Gap filling	✓		
Stumping	✓		
Intercropping	✓		
Soil conservation structures		✓	
Chemical weed control			✓

APPENDIX III: PICTURES DURING DATA COLLECTION